

Supporting Information

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SI Text

Preprocessing of the Data. The database we consider consists of e-mail records for 3,188 e-mail accounts at a university covering an 83-day period (1). Each record comprises a sender identifier, a recipient identifier, the size of the e-mail, and a time stamp with a precision of 1 second. Here, we note some important features of the data that impact our analysis of human activity patterns.

The first important fact is that the data were gathered at an e-mail server, not from the e-mail clients of individual users. It is quite possible that some users have e-mail clients, like Microsoft Outlook, which permit users to send multiple e-mails at once regardless of when the e-mails were composed. The data then appear to show several e-mails sent within a very short time. Moreover, a message to several individuals may be parsed by the server into several e-mails with smaller recipient lists (2). For this reason, single e-mails to multiple recipients were occasionally recorded in the server as multiple e-mails sent in rapid succession. Both the client-side and server-side uncertainties introduce artifacts in the data that may artificially increase the number of e-mails sent within very short time intervals.

To minimize these uncertainties, we preprocessed the data to focus on e-mails that were actually sent by a human. First, we identify sets of e-mails sent by a user that have the exact same size but whose time stamp differs by at most 5 s, which corresponds with the minimal bound on humanly possible inter-event times (Fig. S1A). We then remove all but the first e-mail from the time series of e-mails sent while adjusting the list of recipients of the first e-mail to include all recipients in the parsed e-mails.*

An additional important fact to note is that some of the e-mail accounts do not belong to “typical” users. For example, User 1962 sent only 5 e-mails while receiving 2,284 e-mails. This individual’s e-mail use is too infrequent to provide useful information on human dynamics. Meanwhile User 4099 sent 9,431 e-mails while receiving 0 e-mails. Although it cannot be confirmed because of the anonymous nature of the data, this e-mail account was, in all likelihood, used for bulk e-mails, implying that it cannot provide information on human e-mail usage.

To avoid having our analysis distorted by potentially nonhuman behavior, we first restrict our attention to users who sent at least 1 e-mail every other day, or 41 e-mails over the 83-day experiment (Fig. S1B). Our reasoning is that users sending fewer e-mails do not use e-mail regularly enough to allow us to truly infer patterns of human activity. This procedure excludes 2,619 of the 3,188 original e-mail accounts.

Next, we examine the ratio of the number of e-mails received to the number of e-mails sent to determine what constitutes a “typical” user. The distribution of the log of this ratio has central tendencies (Fig. 1C), and we use this fact to consider only those users in our study who are within 2.5 standard deviations from the mean. This added constraint excludes 165 additional users.

Finally, we exclude the final 10 users due to the computational costs of our simulated annealing procedure. We thus focus here on the 394 users who fulfill the above criteria.

Null Model. The truncated power-law null model is based on the assumption that the time series is constructed by randomly drawing inter-event times from a truncated power-law distribution with exponent $\alpha = 1$:

$$p(\tau|\tau_{\min}, \tau_{\max}) = \begin{cases} 0 & \text{if } \tau < \tau_{\min} \\ \tau^{-\alpha} e^{-\tau/\tau_{\max}} & \text{if } \tau \geq \tau_{\min} \end{cases} \quad [1]$$

where $\tau_{\min}$ is the lower-bound cutoff on the power-law scaling, $\tau_{\max}$ is the exponential truncation, and Γ is the upper incomplete gamma function.

Just as with the cascading nonhomogeneous Poisson process, we must estimate the unknown parameters $\theta = \{\tau_{\min}, \tau_{\max}\}$ of the truncated power-law model. Typically researchers use maximum-likelihood estimation, which maximizes the log-likelihood function $\ell(\theta) = \sum \ln p(\tau_i|\theta)$, to identify the best-estimate parameters $\hat{\theta}$ for a model. In most cases, maximum-likelihood estimation provides an asymptotically unbiased estimate of the parameters $\hat{\theta}$. Estimating the parameters for the truncated power-law model, however, is complicated by the fact that the log-likelihood function is finite only when $\hat{\tau}_{\min} \leq \min\{\tau_i\}$; that is, maximum-likelihood estimation provides a biased estimate of $\tau_{\min}$ (Fig. S3), and the maximum-likelihood prediction of the truncated power-law model does not visually capture the asymptotic scaling of the empirically observed inter-event time distribution (Fig. S4).

To circumvent these problems, we use least-area estimation, which minimizes the area test statistic between the cumulative distribution $P(\tau)$ and the truncated power-law’s prediction $P(\tau|\theta)$. Importantly, least-area estimation is asymptotically unbiased (Fig. S3), and it also provides an accurate description of the asymptotic scaling of the empirical inter-event time distribution (Fig. S4). Although least-squares estimation also provides an asymptotically unbiased estimate of the parameters of the truncated power-law model, we use least-area estimation so that our parameter-estimation method is compatible with our parameter-estimation method for the cascading nonhomogeneous Poisson process, which also uses least-area estimation.

In Fig. S6, we present the best-estimate predictions and Monte Carlo hypothesis-testing results for the truncated power-law model for each of the 394 users under consideration.

Additional Evidence for a Homogeneous Poisson Cascade. In the main text, we argued that the N_a additional events that occur during active interval cascades are governed by a homogeneous Poisson process with a constant rate ρ_a . One could simply justify the choice of a homogeneous Poisson process during active intervals on the basis that (i) it is typically used to model human activity, and (ii) it is one of the simplest possible stochastic models. We go beyond these historical and Occam’s Razor arguments, and we offer quantitative supporting evidence for the existence of a homogeneous Poisson process during active intervals.

Previously, we have used our simulated annealing procedure to find the optimal configuration of active intervals such that the empirical inter-event time distribution matches the predictions of the cascading nonhomogeneous Poisson process. Here, we modify our simulated annealing procedure to extract the nature of the inter-event time distribution, given that a nonhomoge-

*A more aggressive preprocessing method would also attempt to remove blind-carbon-copied (BCC) e-mails. The basic idea is that an e-mail with BCC recipients will have its size increased by a few bytes because of the addition of outgoing headers. E-mails to the BCC recipients would be sent by the server shortly after the e-mail to visible recipients and would thus increase the number of very small inter-event times. We chose to err on the side of caution and not attempt to remove e-mails with BCC recipients, because their detection is more subjective.

following experiment. First, we generate a synthetic dataset from a cascading nonhomogeneous Poisson process with parameters θ , recording the actual active interval configuration C_o from which we compute the actual parameters θ_o . Then we use our simulated annealing procedure to identify the best-estimate active interval configuration $\hat{C}$ from which we calculate the estimated parameters $\hat{\theta}$. Finally, we compare the original active interval configuration C_o with the estimated active interval configuration $\hat{C}$ and the original parameters θ_o with the estimated parameters $\hat{\theta}$. We repeat this procedure with 1,000 synthetic datasets generated from parameters θ similar to the parameters that we estimate for the users in the data. This procedure allows us to assess how accurate our best-estimate active interval configuration $\hat{C}$ is to the optimal active interval configuration C_o and whether the estimated parameters $\hat{\theta}$ are statistically similar to the actual parameters θ_o .

Because the parameters of the cascading nonhomogeneous Poisson process are calculated directly from the active interval configuration $\hat{C}$, the most stringent assessment of the parameter-estimation procedure is to determine how accurately the active intervals are identified. As a reminder, every pair of consecutive events can either be merged into the same active interval or split into separate active intervals during the simulated annealing procedure. Thus, we can measure the accuracy of the estimated active interval configuration $\hat{C}$ by determining the fraction of consecutive events that are correctly merged into the same active interval or split into separate active intervals when compared with the actual active interval configuration C_o (5). If our estimation procedure randomly merged or split consecutive events, we would expect an accuracy of 50%—the worst-case

scenario. In fact, our simulated annealing procedure has an average accuracy of 90.3%, noticeably outperforming the random active interval assignment (Fig. S5A).

The question is whether an accuracy of 90.3% is enough to produce best-estimate parameters $\hat{\theta}$ that are statistically similar to the actual parameters θ_o . For the scalar parameters in our model N_w and ρ_a , this is relatively straightforward. In the case of the number of active intervals per week N_w , for example, we can look at the scaled residuals $(\hat{N}_w - N_{wo})/N_{wo}$ between the estimated number of active intervals per week $\hat{N}_w$ and the number of active intervals per week that were used to generate the synthetic data N_{wo} . As we would expect if the simulated annealing procedure were accurate, the distribution of scaled residuals is highly peaked around zero (Fig. S5B). We demonstrate the same finding with the distribution of scaled residuals for the homogeneous Poisson rate ρ_a during active intervals (Fig. S5C).

We evaluate the statistical similarity for the distribution parameters of our model $p_w(t)$, $p_d(t)$, and $p(N_a)$ in a slightly different way. Here, we use the AUC test statistic (5) to determine whether the estimated distribution $\hat{p}(\cdot)$ is similar to the actual distribution $p_o(\cdot)$. The AUC test statistic quantifies the probability that a random variate drawn from $\hat{p}(\cdot)$ will be greater than a random variate drawn from $p_o(\cdot)$. If the distribution $\hat{p}(\cdot)$ estimated by our simulated annealing procedure is accurate, we would expect that $\hat{p}(\cdot)$ is similar to $p_o(\cdot)$, which is characterized by an AUC that approaches 0.5. We demonstrate that the distribution of AUC test statistics is highly peaked, indicating that our simulated annealing procedure provides accurate estimates of the parameters of the cascading nonhomogeneous Poisson process (Fig. S5 D–F).

1. Eckman J-P, Moses E, Sergi D (2004) Entropy of dialogues creates coherent structure in e-mail traffic. *Proc Natl Acad Sci USA* 101:14333.
2. Berson A (1992) *Client/Server Architecture* (McGraw-Hill, New York).
3. D'Agostino RB, Stephens MA (1986) *Goodness-of-Fit Techniques* (Marcel Dekker, New York).

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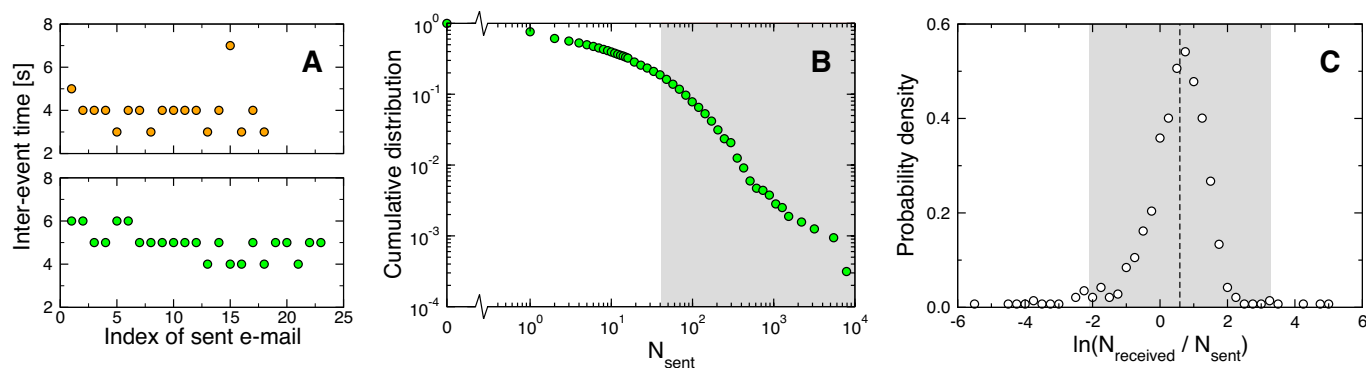


Fig. S1. Preprocessing of the data. (A) Limits on time required to write and send an e-mail. Two e-mail users from our research group sent several empty e-mails as rapidly as possible without using hot keys. Bear in mind that this time interval does not even include the time required to actually compose an outgoing message. We find that it takes at least 3–7 seconds just to send an empty e-mail, and we use this information to identify unique messages from the data. (B) The cumulative distribution of the number of e-mails sent by the 3,188 users over 83 days. To avoid characterizing users who use e-mail infrequently, we consider the 569 users who sent at least 41 e-mails over 83-day experiment (shaded region). Note that this removes the 759 users who sent no e-mails and the 370 users who sent 1 e-mail. (C) The distribution of the log of the ratio of the number of e-mails received to the number of e-mails sent. This distribution is highly peaked near the mean (dashed line). We use this fact to develop a criteria to identify typical e-mail users that sent at least 41 e-mails over 83 days and differentiate them from bulk e-mail accounts, listserves, and e-mail accounts that are rarely used by their owners. We keep the 394 users who fall within 2.5 standard deviations of the mean (shaded region).

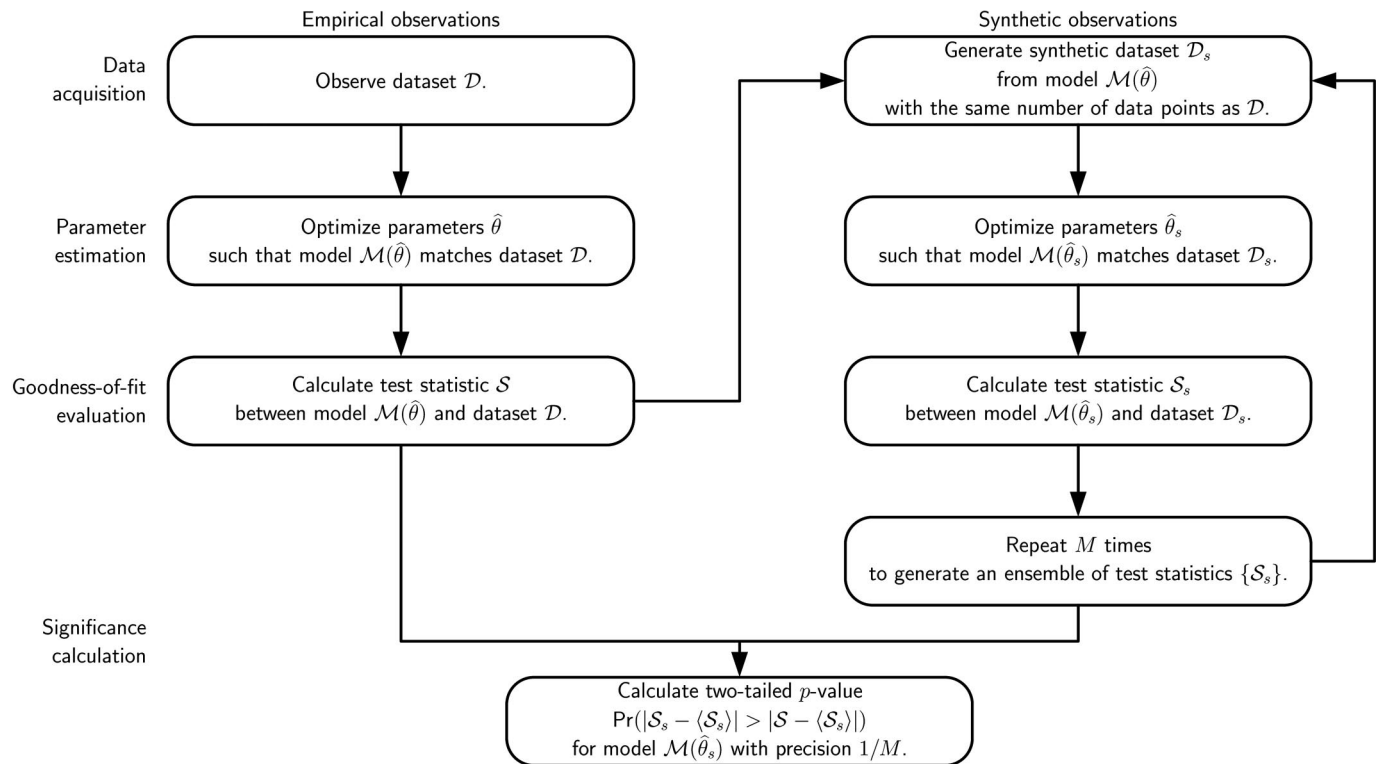


Fig. S2. Monte Carlo technique for determining whether a model $\mathcal{M}(\theta)$ with unknown parameters θ is consistent with observed data $\mathcal{D}$. First, we calculate the best-estimate parameters $\hat{\theta}$ for model $\mathcal{M}$ by optimizing the model's fit to the data $\mathcal{D}$ using the parameter estimation procedures for that model. Second, we compute the test statistic S between the model predictions $P_{\mathcal{M}}(u|\hat{\theta})$ and the empirical data $P_{\mathcal{D}}(u)$. Next, we generate a synthetic dataset $\mathcal{D}_s$ from model $\mathcal{M}(\hat{\theta})$ using the best-estimate parameters $\hat{\theta}$, and we treat the synthetic data exactly the same as we treated the empirical data: First, we calculate the best-estimate $\hat{\theta}_s$ for model $\mathcal{M}$ by optimizing the model's fit to the synthetic data $\mathcal{D}_s$ using the parameter-estimation procedures for that model; second, we compute the test statistic S_s between the model predictions $P_{\mathcal{M}}(u|\hat{\theta}_s)$ and the synthetic data $P_{\mathcal{D}_s}(u)$. We continue generating synthetic datasets until we accumulate an ensemble of M Monte Carlo test statistics $\{S_s\}$. Finally, we calculate a 2-tailed P value with a precision of $1/M$ by computing $\Pr(|S_s - \langle S_s \rangle| > |S - \langle S \rangle|)$. In the case of the cascading nonhomogeneous Poisson process, we estimate the parameters $\{\hat{\theta}\}$ using our simulated annealing procedure (which minimizes the area test statistic) and for the truncated power-law model, we estimate the parameters $\{\hat{\theta}\}$ using least-area estimation. For both models, we use the area test statistic $S = A$ (see *Methods*) as a means for comparing each model $\mathcal{M}$ with the $\mathcal{D}$. As demonstrated by this diagram, the hypothesis-testing procedure implicitly relies on the accurate estimation of the parameters θ for model $\mathcal{M}$. It is therefore critical to validate novel parameter estimation techniques, such as simulated annealing, to ensure that the hypothesis testing procedure is valid.

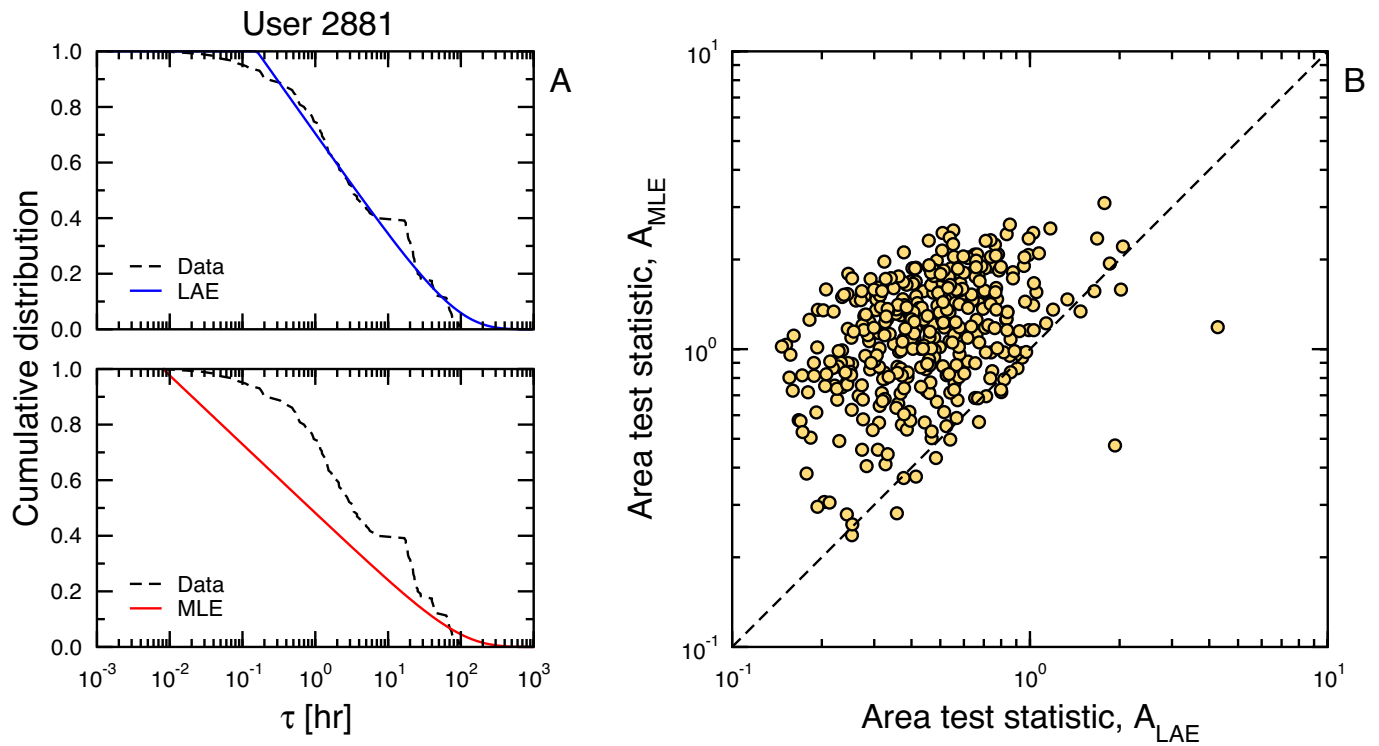
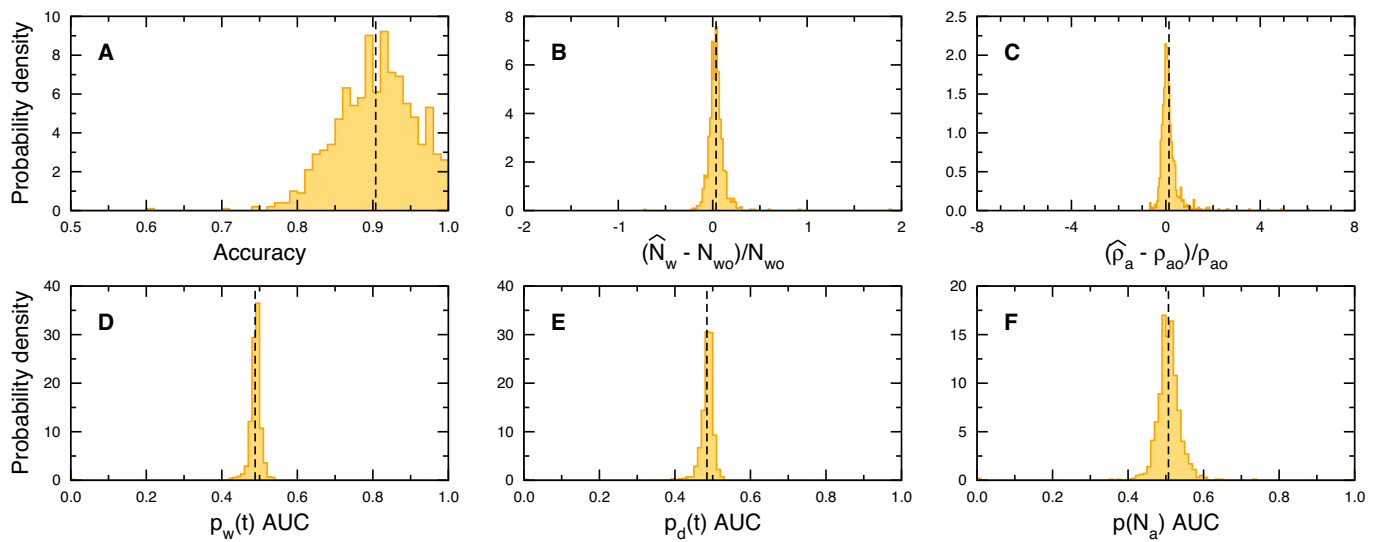


Fig. 54. Comparison of estimating the truncated power-law null model parameters by using least-area estimation or maximum-likelihood estimation (MLE) for empirical datasets. (A) Least-area parameter estimation (*Upper*, blue) accurately depicts the asymptotic scaling of the empirical inter-event time distribution. By comparison, maximum-likelihood estimation (*Lower*, red) does not visually depict the asymptotic scaling of the empirical inter-event time distribution because the maximum-likelihood estimate of $\tau_{\min}$ is biased because $\tau_{\min} \leq \min\{\tau_i\}$. (B) Scatter plot of the area test statistic A from maximum-likelihood estimation (MLE) and least-area estimation (LAE) for each of the 394 users under consideration. The MLE area test statistic A_{MLE} is systematically larger than the LAE area test statistic A_{LAE} for nearly all users under consideration, indicating that maximum-likelihood does not capture the asymptotic scaling of the empirically observed inter-event time distributions for our truncated power-law null model.



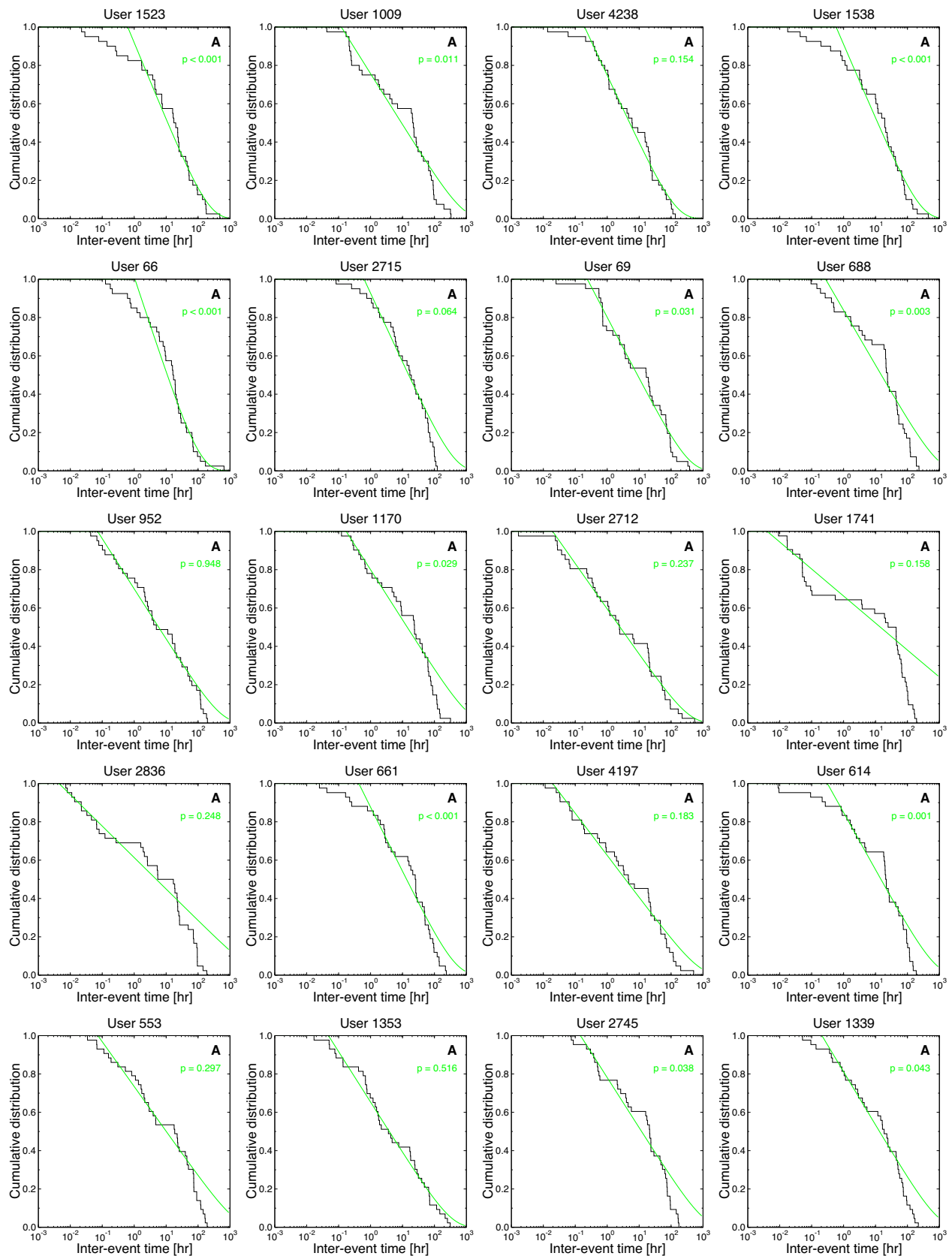


Fig. S6. Comparison of the predictions of the truncated power-law null model with the empirical data for each of the 394 users under consideration. (A) Cumulative distribution of inter-event times $P(\tau)$. We use the area test statistic A and Monte Carlo techniques to quantify the significance of the agreement between the model and the data. As these figures are presented, the area test statistic A is the area between the 2 curves. The P value is computed by using our Monte Carlo hypothesis testing procedure (see *Monte Carlo Hypothesis Testing* in [SI Text](#)).

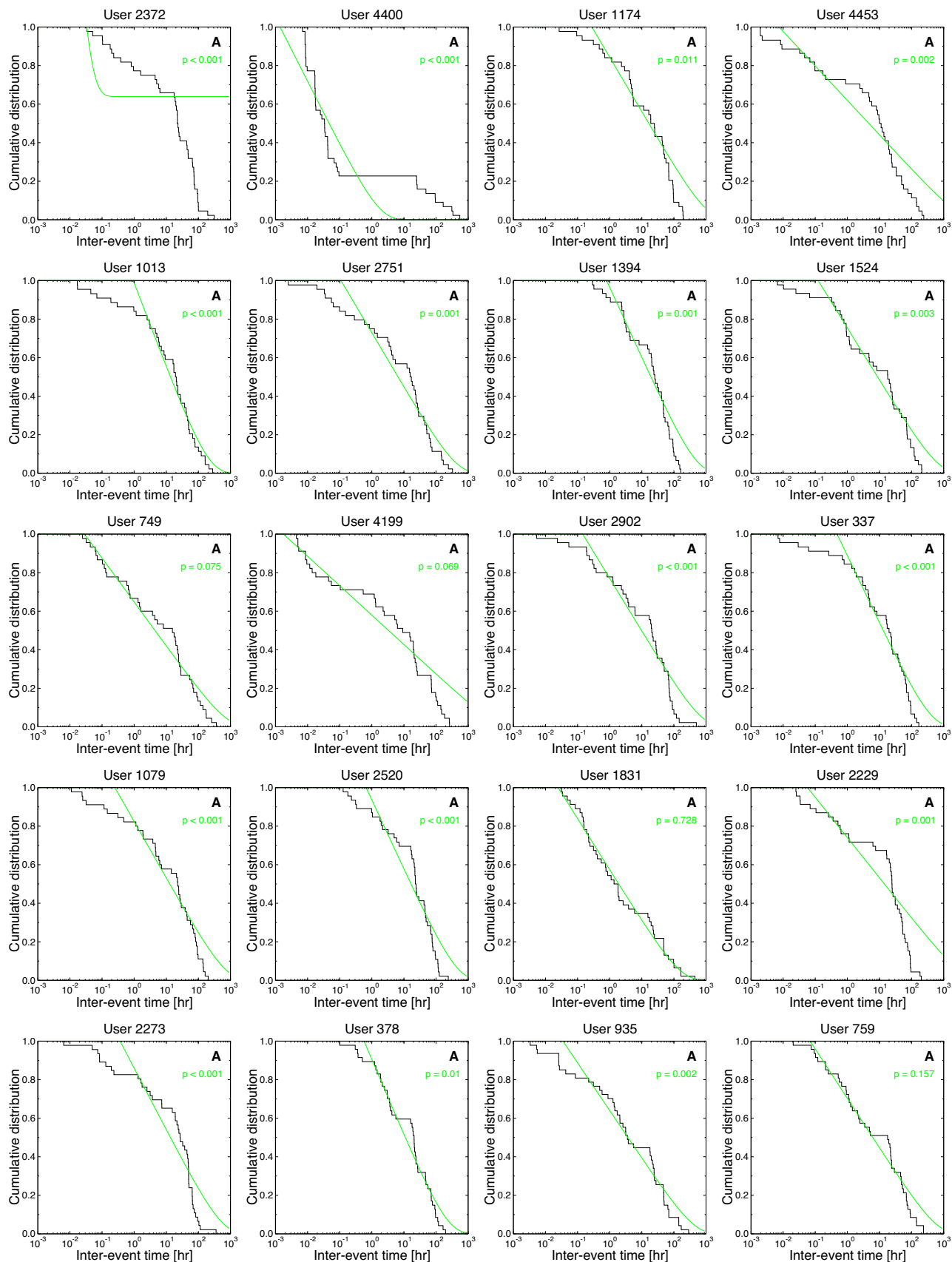


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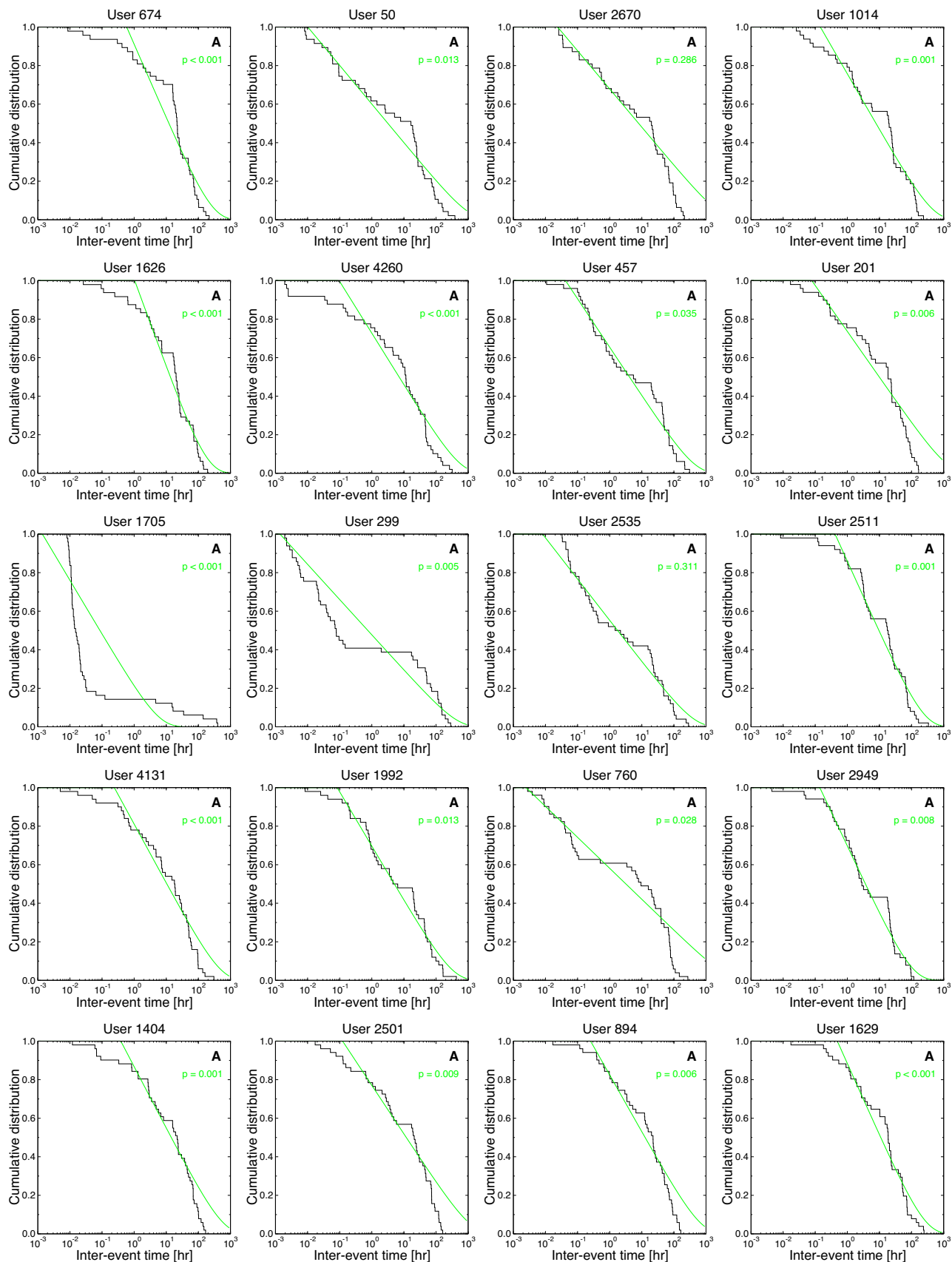


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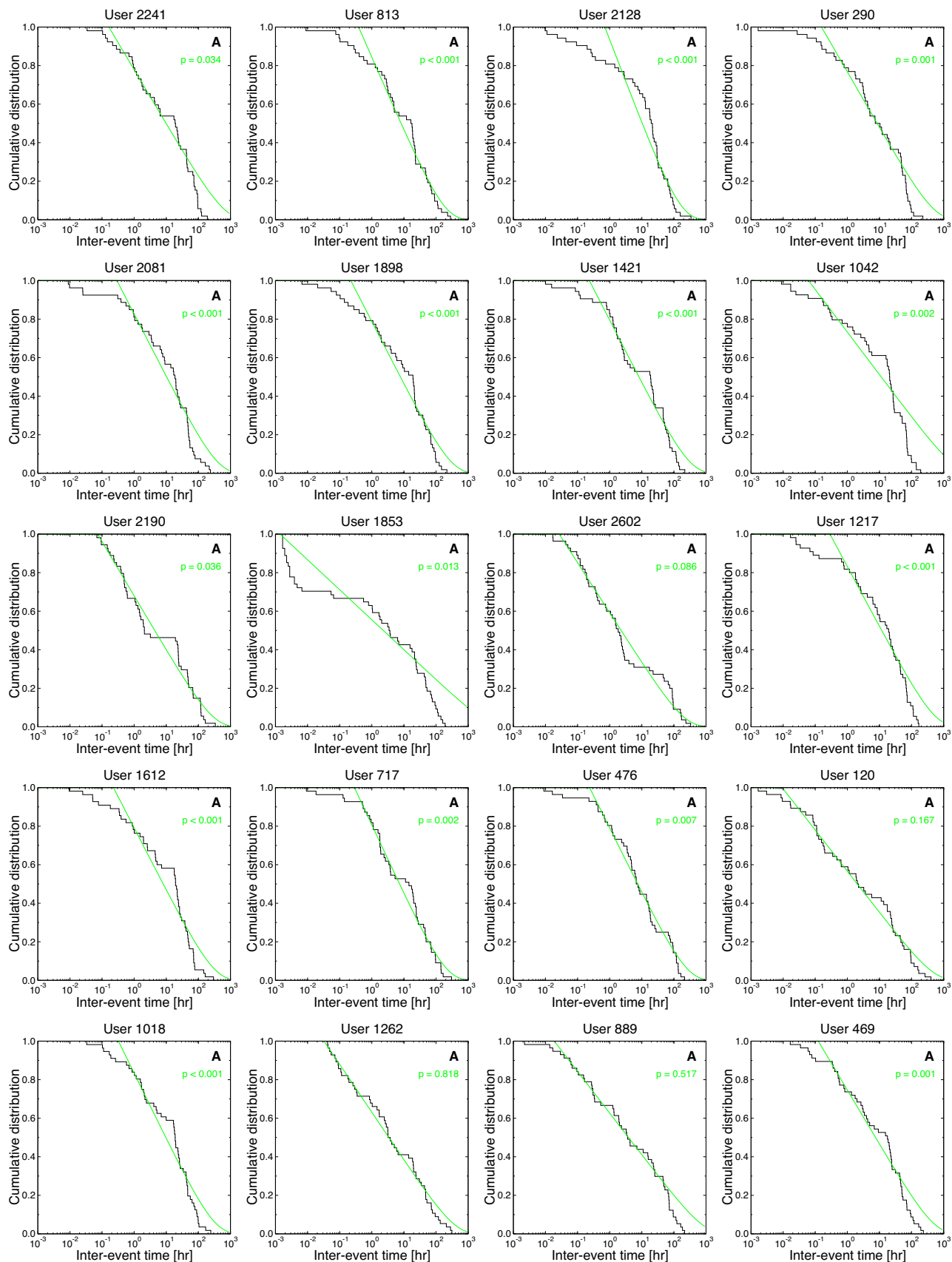


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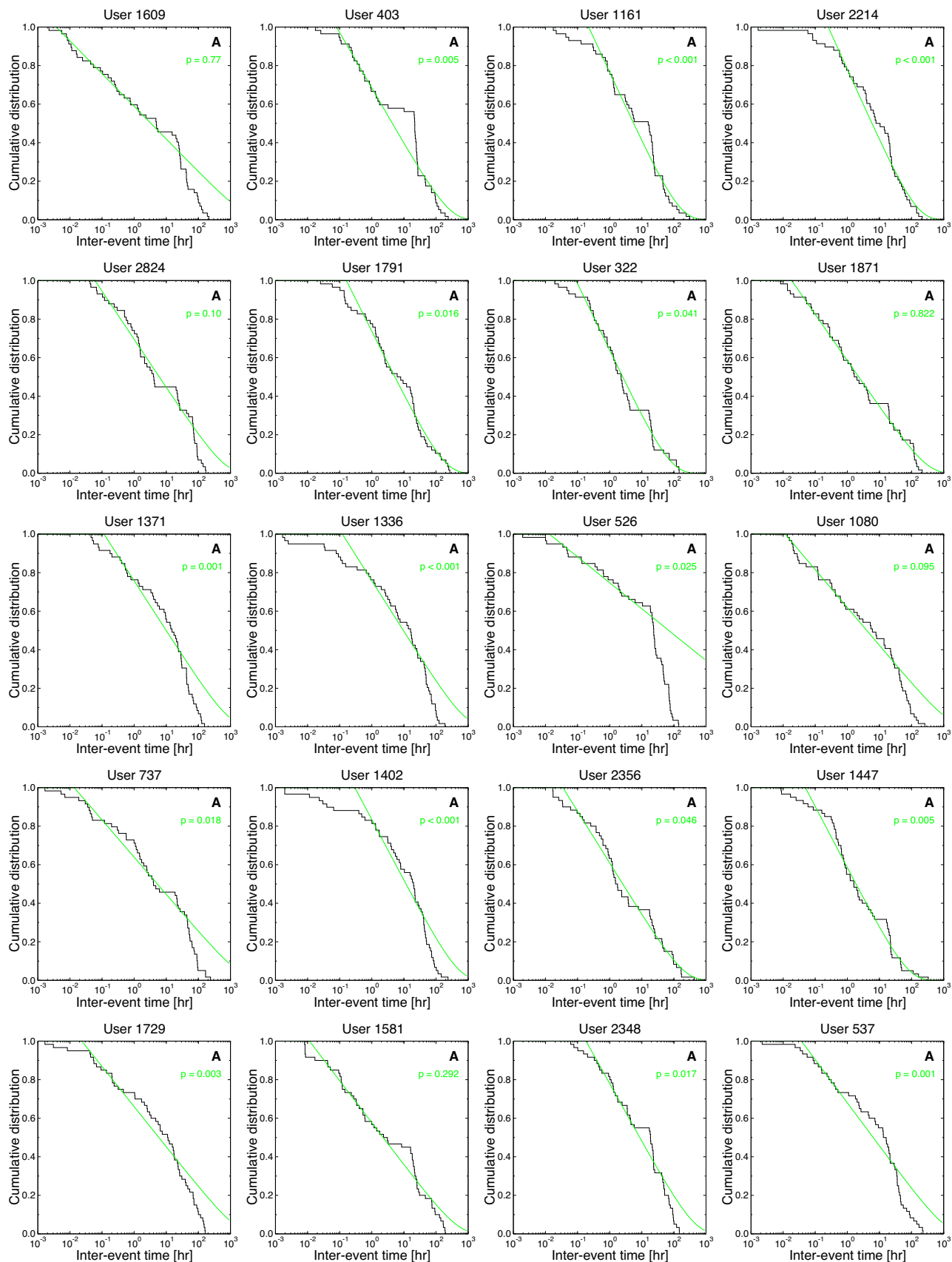


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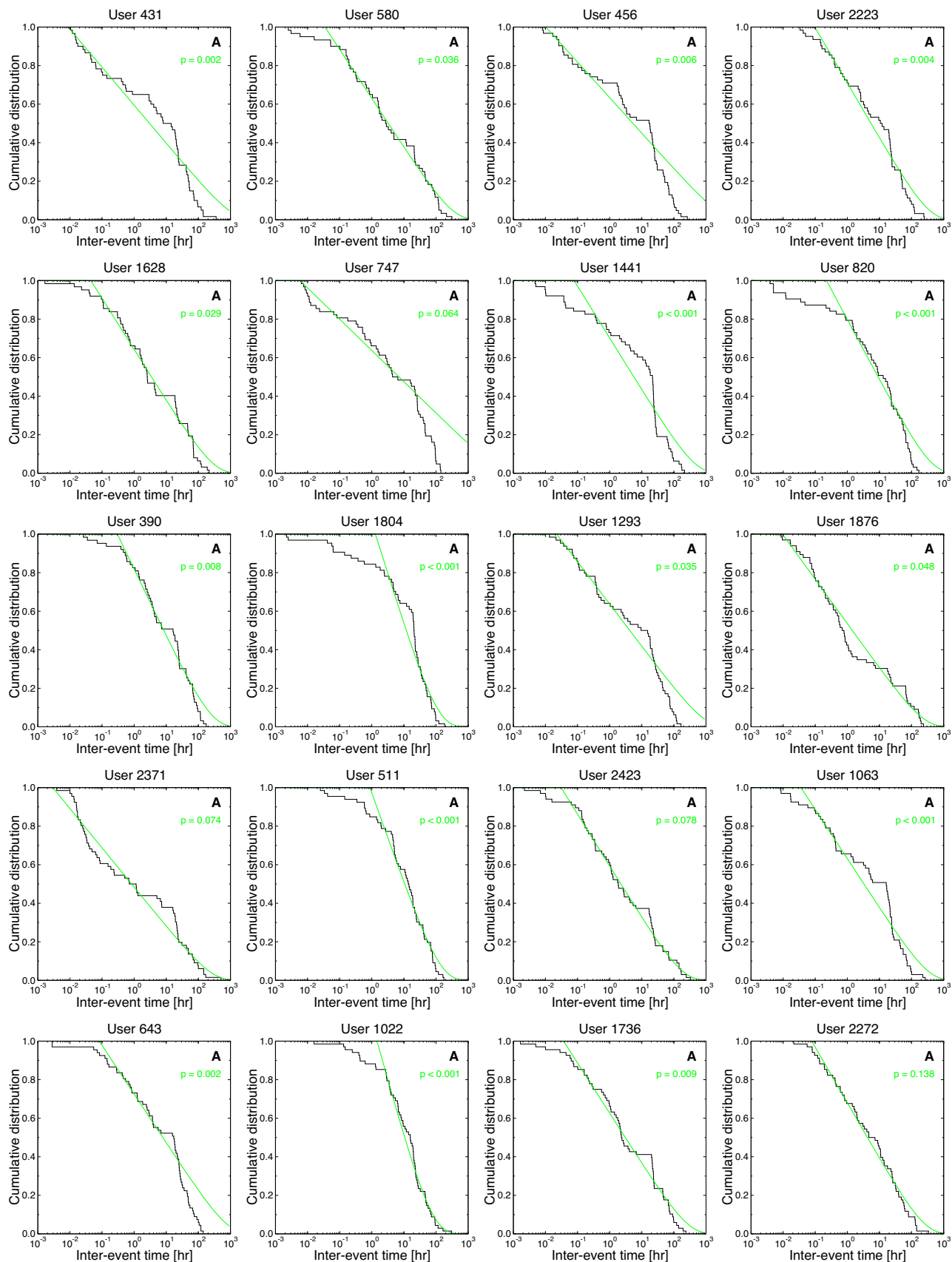


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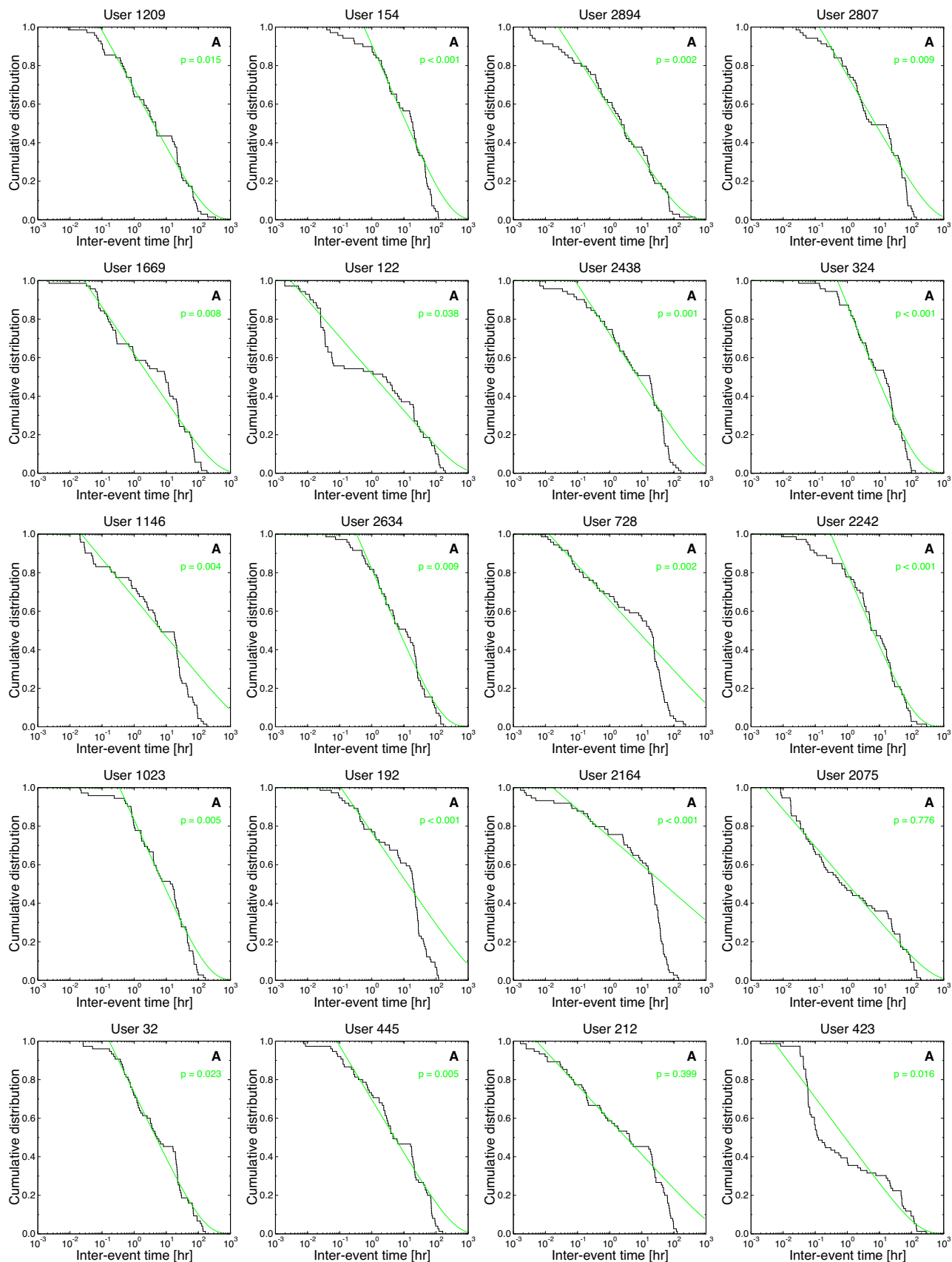


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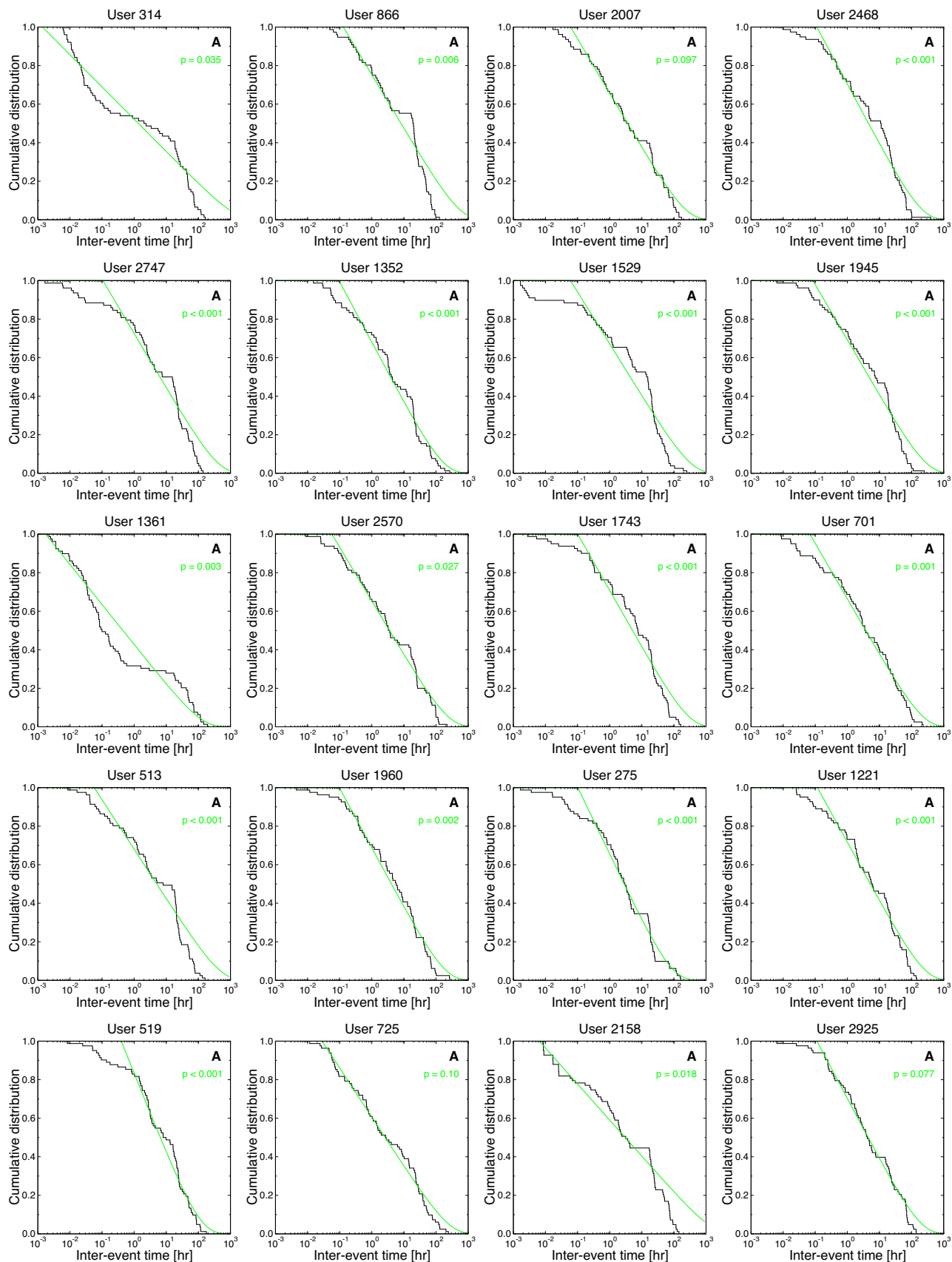


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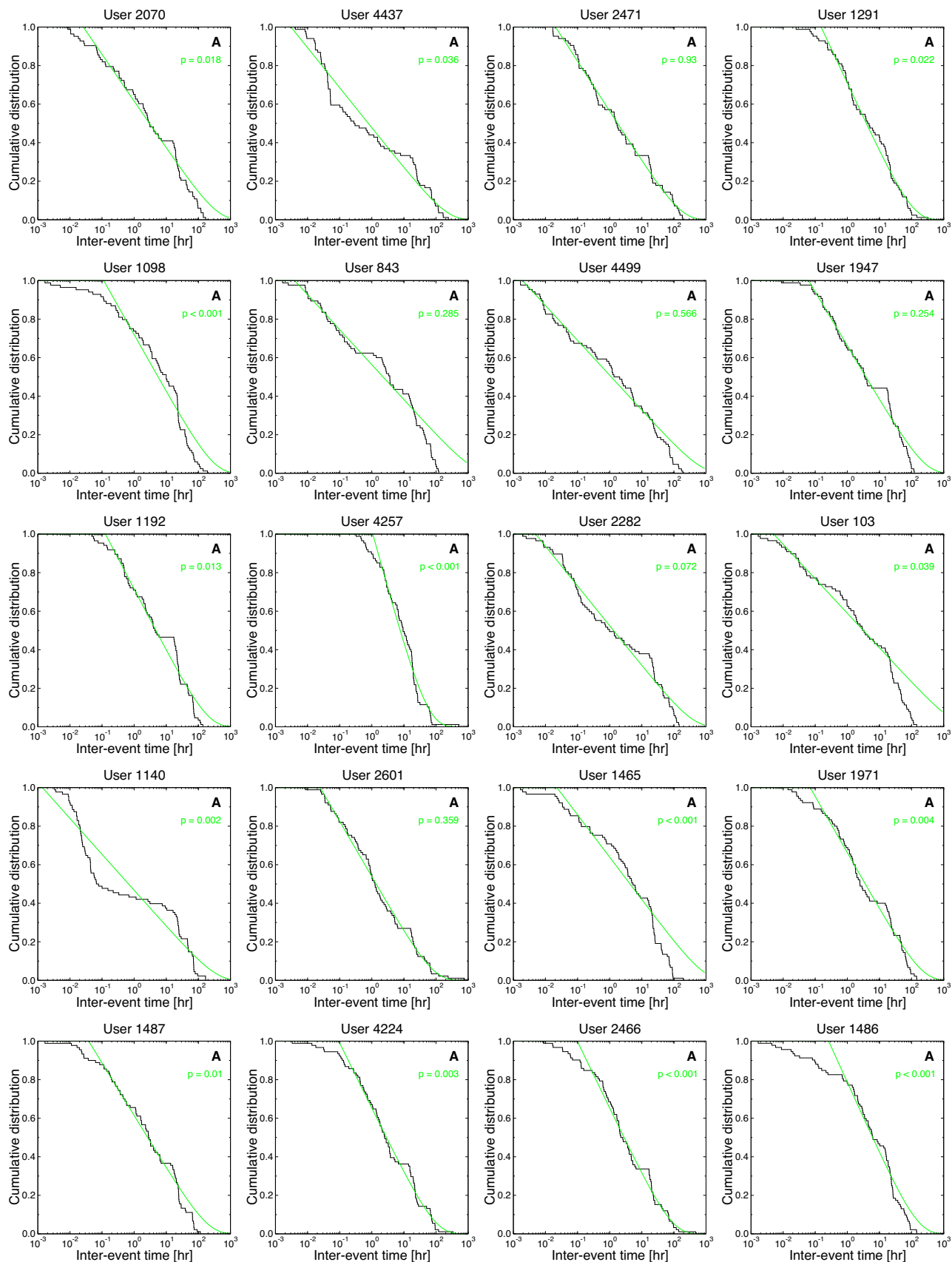


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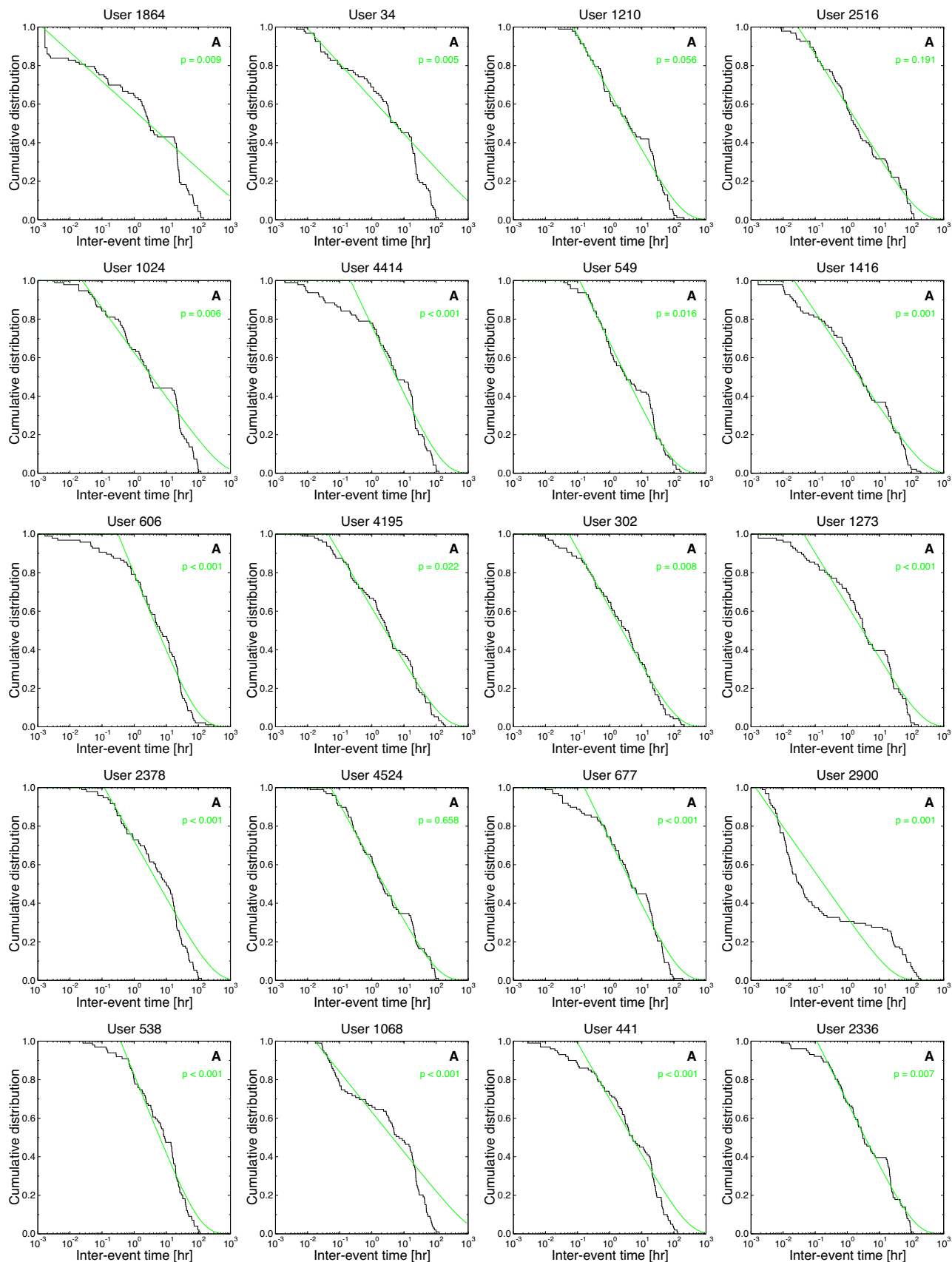


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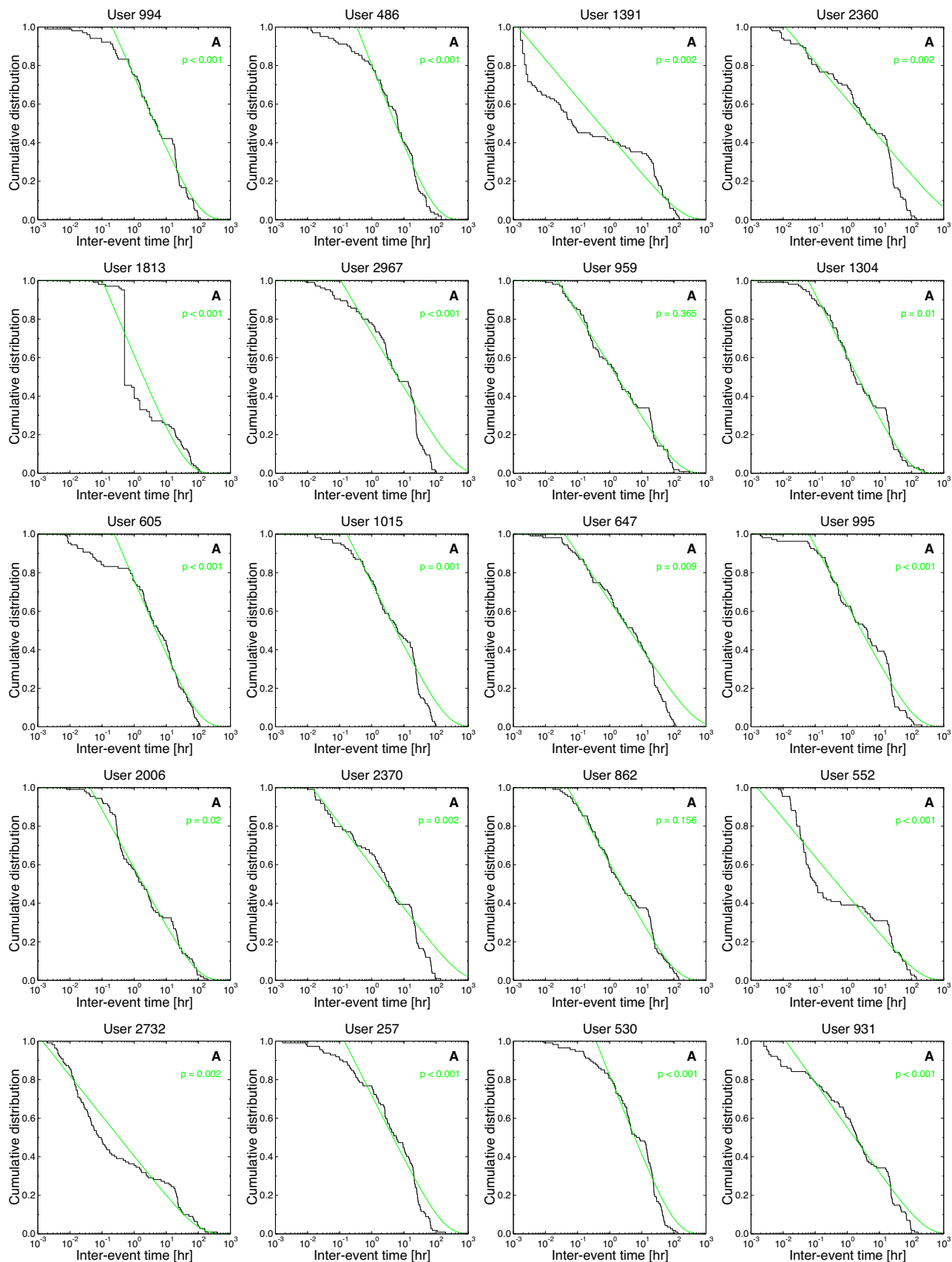


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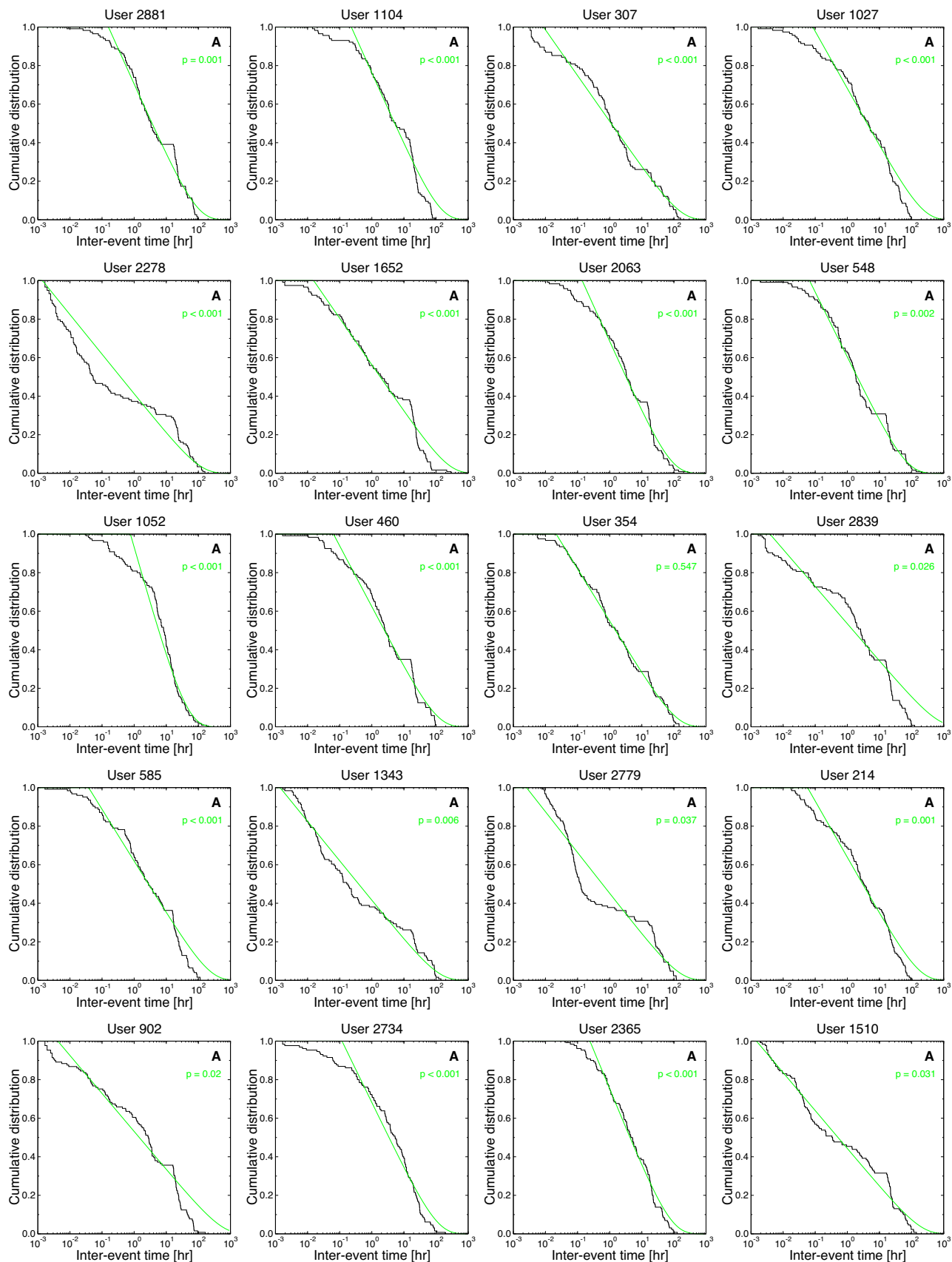


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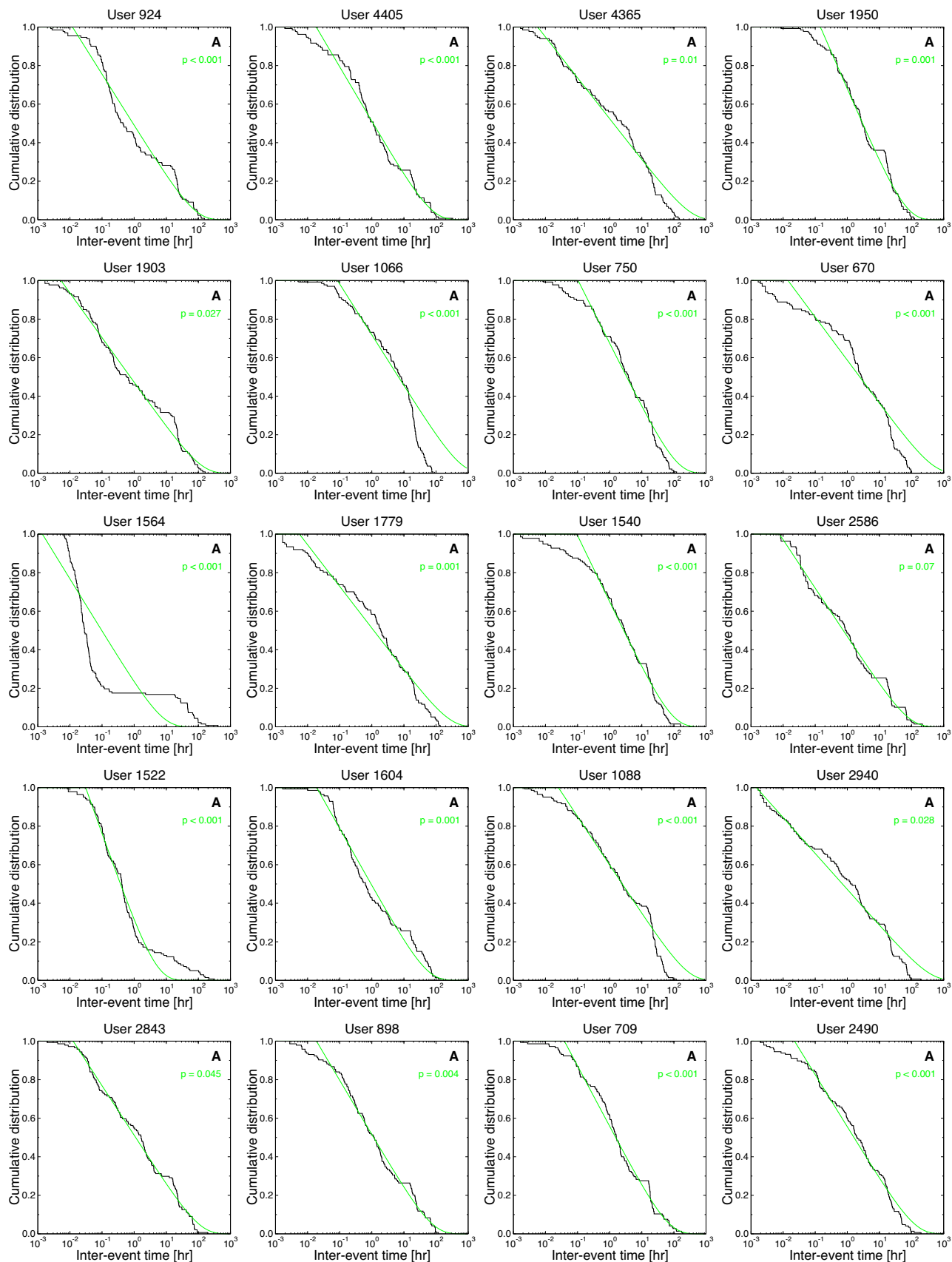


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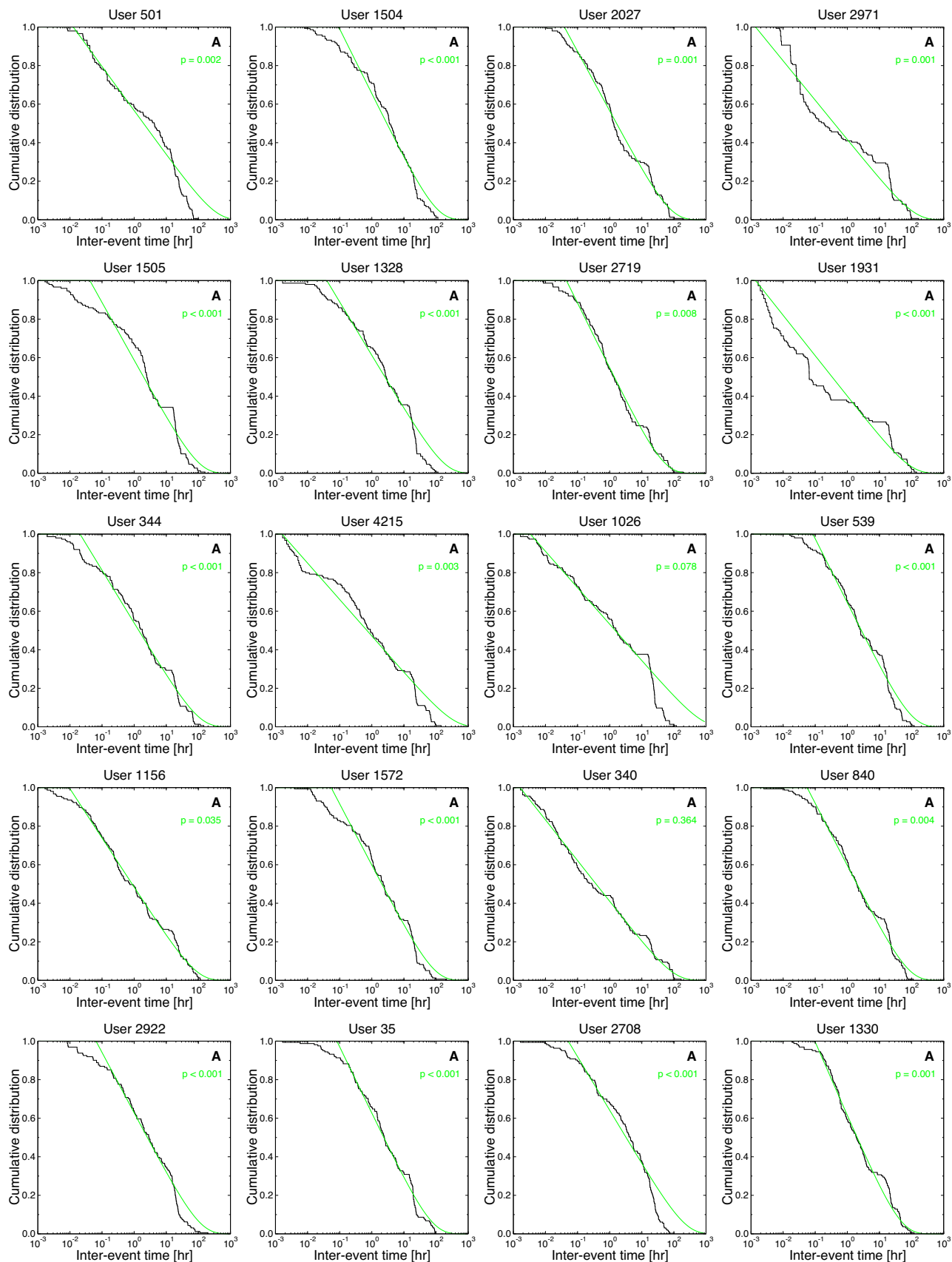


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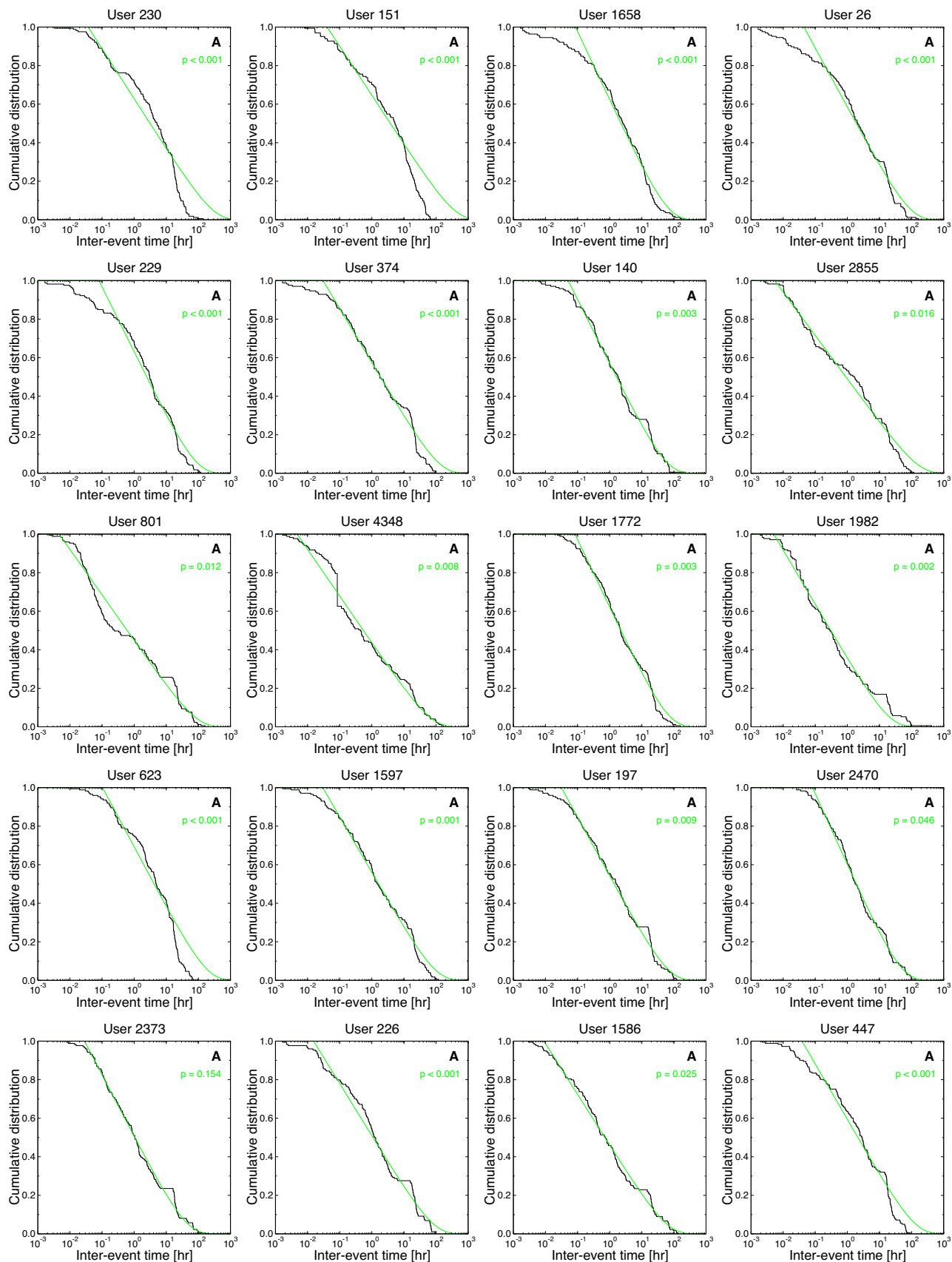


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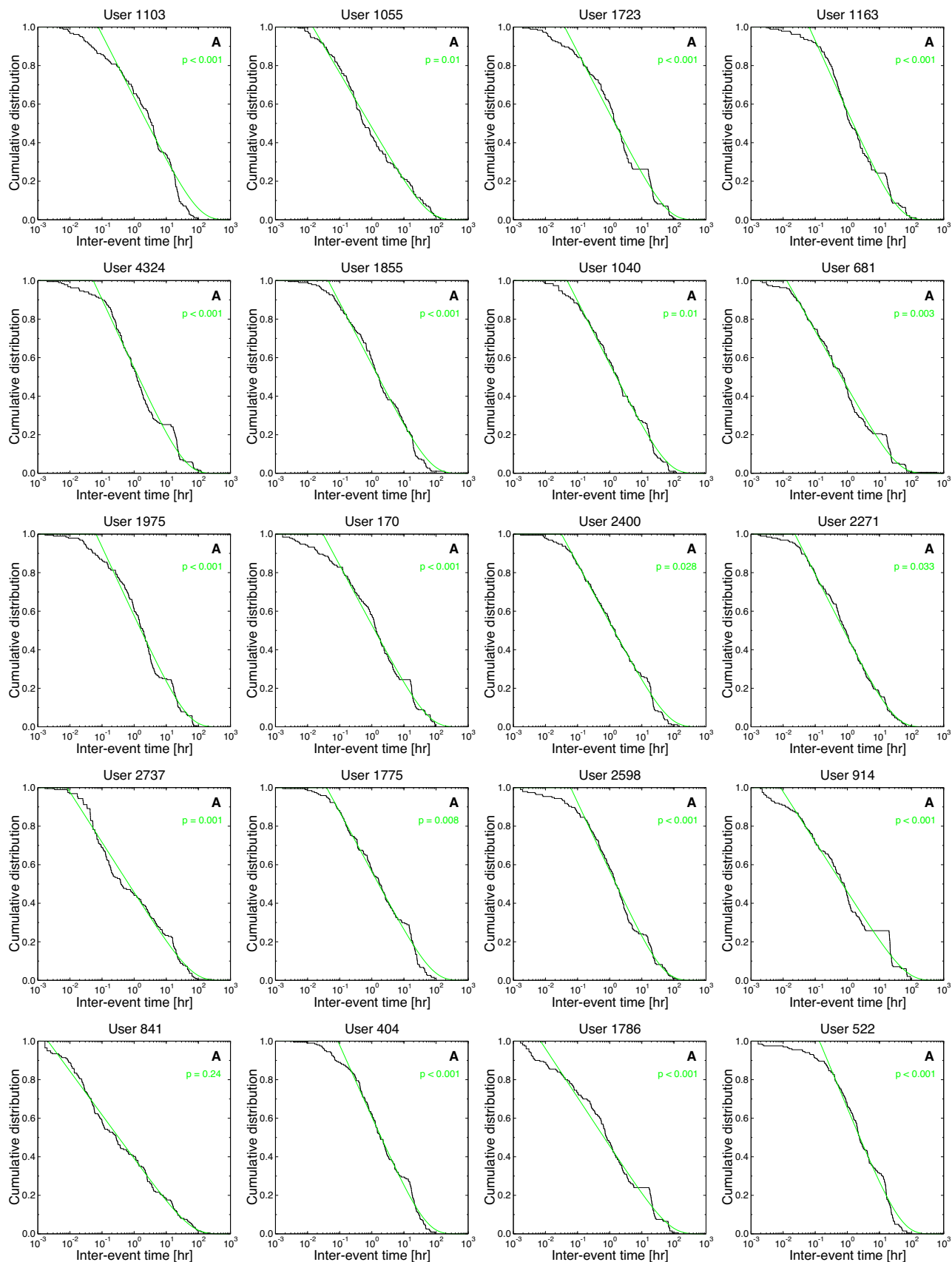


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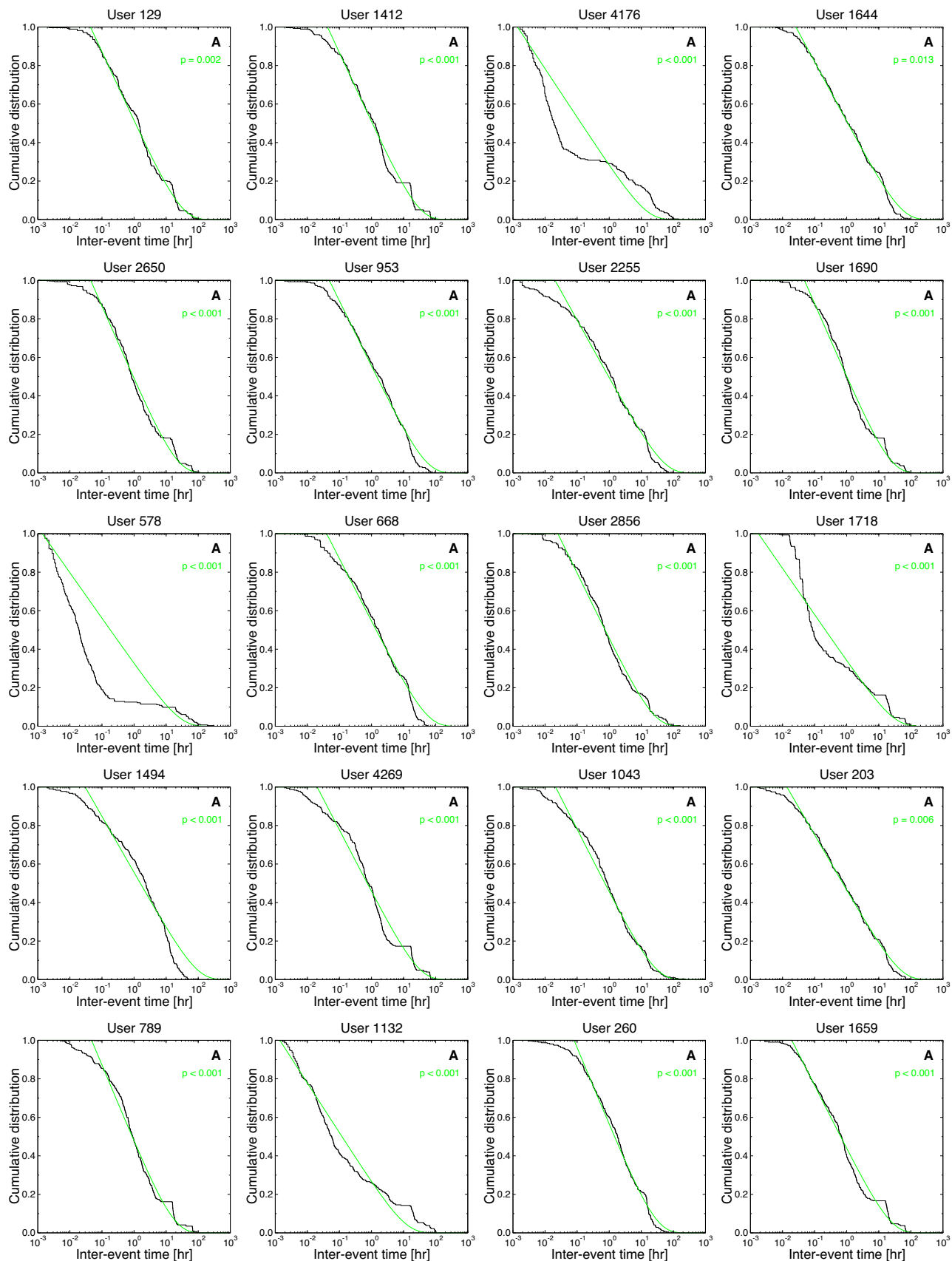


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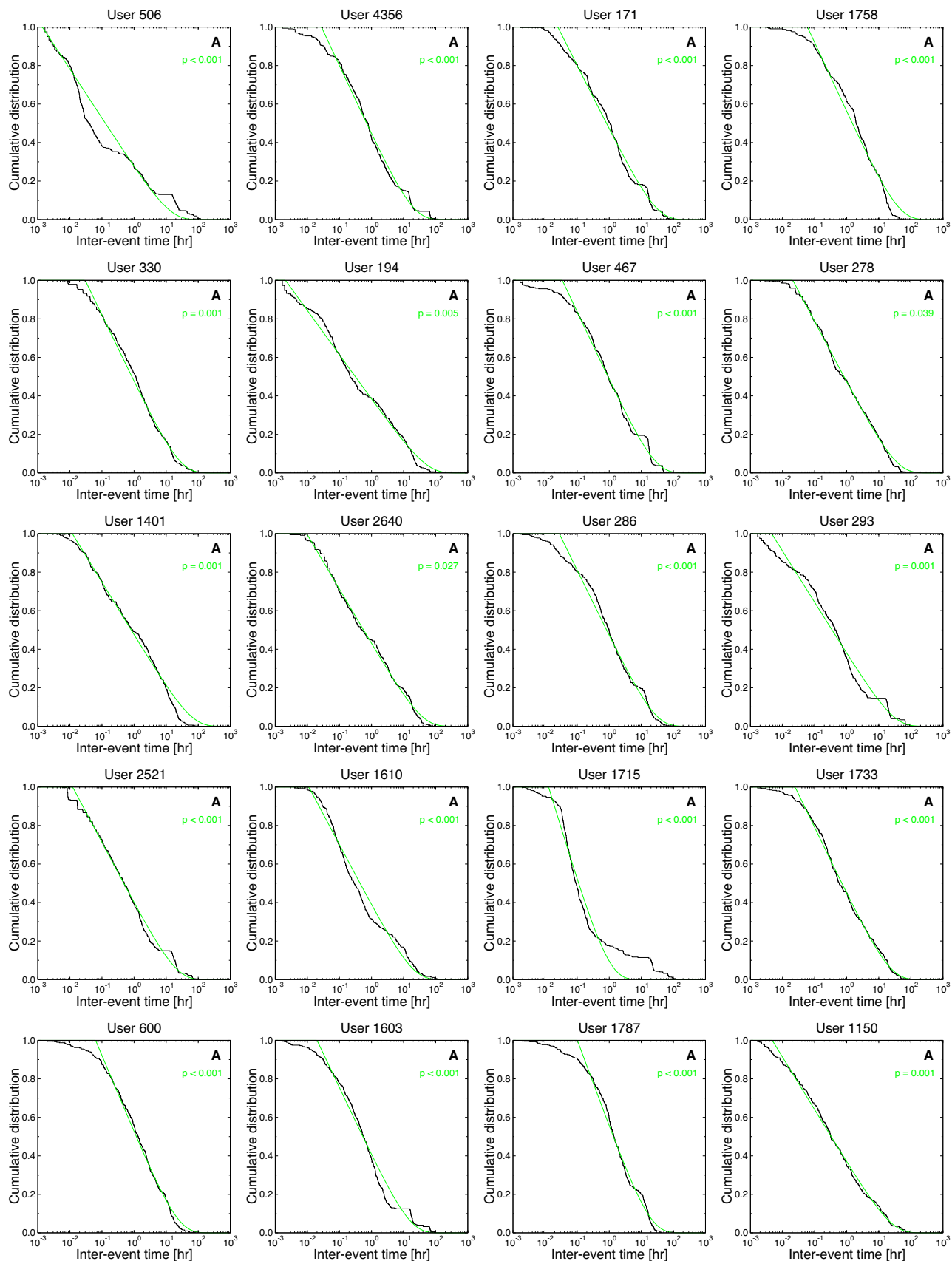


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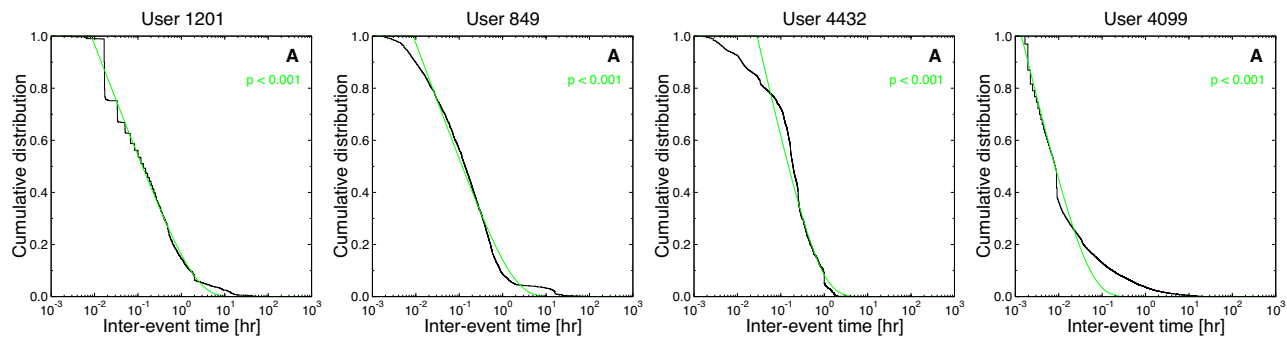


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Fig. S7. The cumulative distribution of inter-event times during active intervals τ_a for 379 of the 394 users under consideration listed in order of increasing e-mail activity. We select the 379 users who sent less than 5 e-mails per day for computational reasons. We identify active intervals by modifying our simulated annealing procedure to optimize the agreement between the nonhomogeneous Poisson process and the cumulative distribution of τ_b —the time between the start of 2 consecutive active intervals. Each image displays the empirical distribution of inter-event times during active intervals τ_a (black). We estimate the maximum-likelihood best-fit exponential distribution $P(\tau_a) = e^{-\rho \tau_a}$ (blue) as a guide to the eye. The qualitative agreement between the empirical distribution of inter-event times during active intervals and the best-fit exponential supports the claim that active intervals are indeed governed by a homogeneous Poisson process. One difference between the simulated annealing to identify the parameters of the cascading nonhomogeneous Poisson process and the simulated annealing method used here is that we consider merging all consecutive events into active intervals and not only those consecutive events that have $\tau < 0.1/\langle \rho(t) \rangle$. As a result, unphysically large inter-event times τ are oftentimes grouped into active intervals.

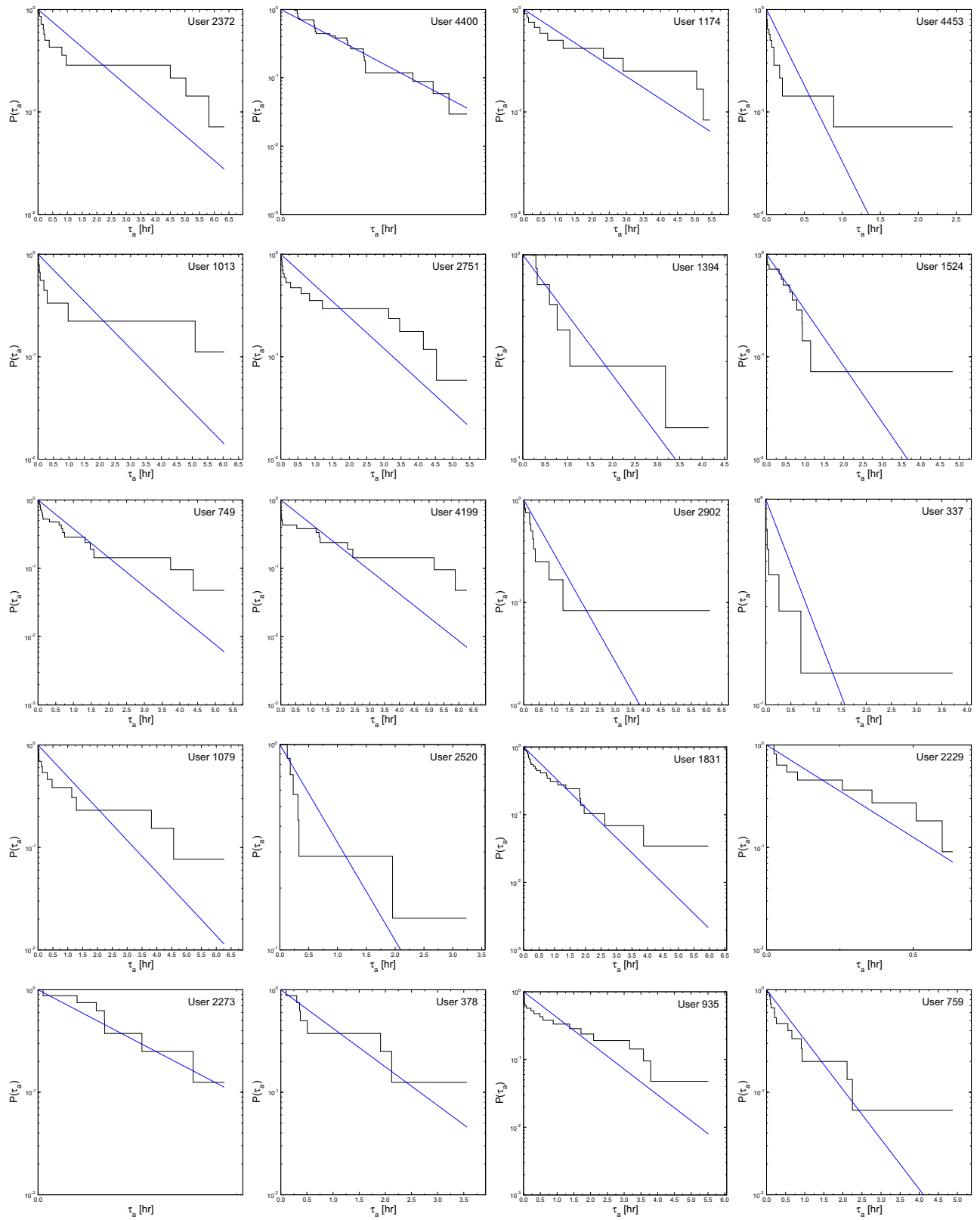


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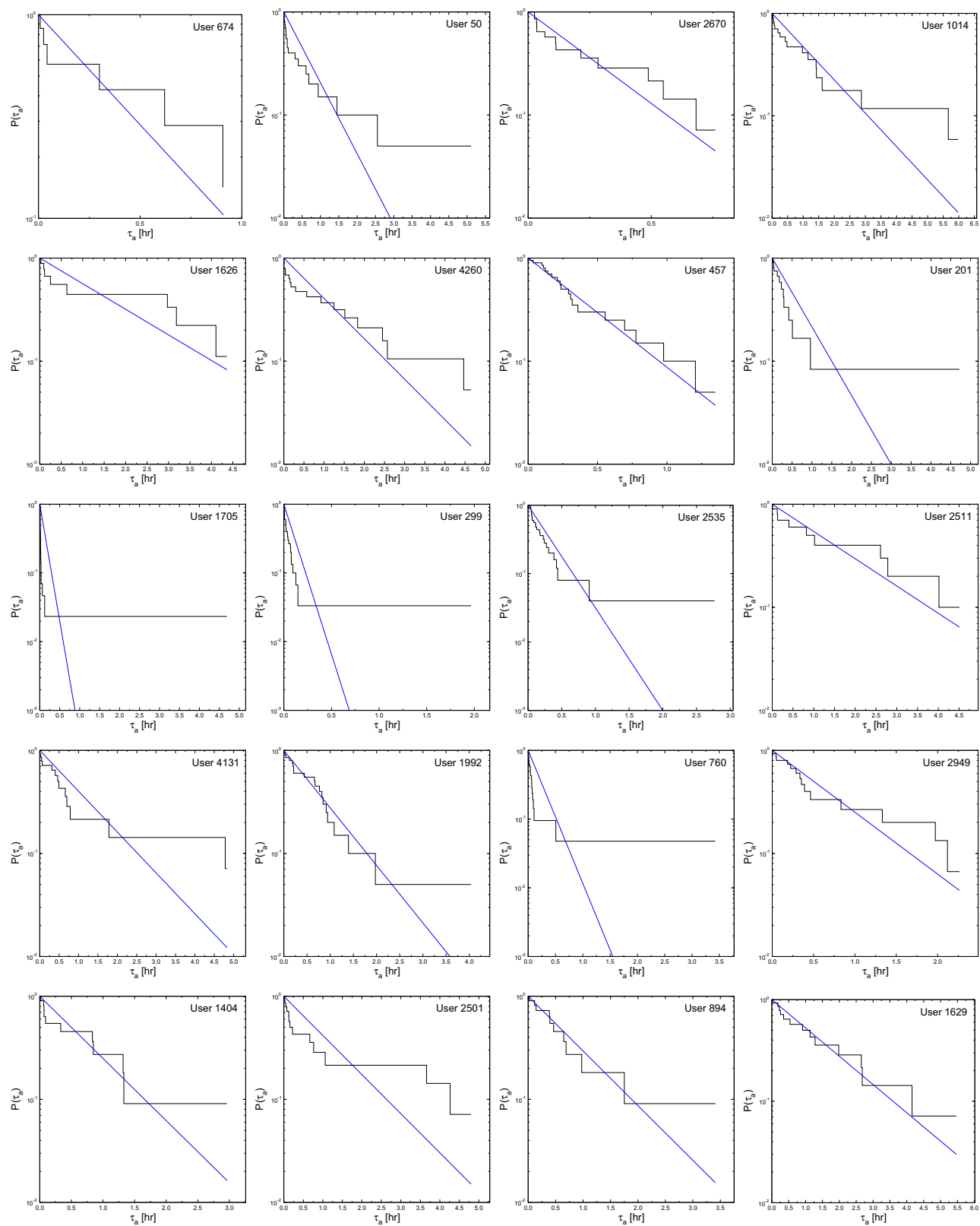


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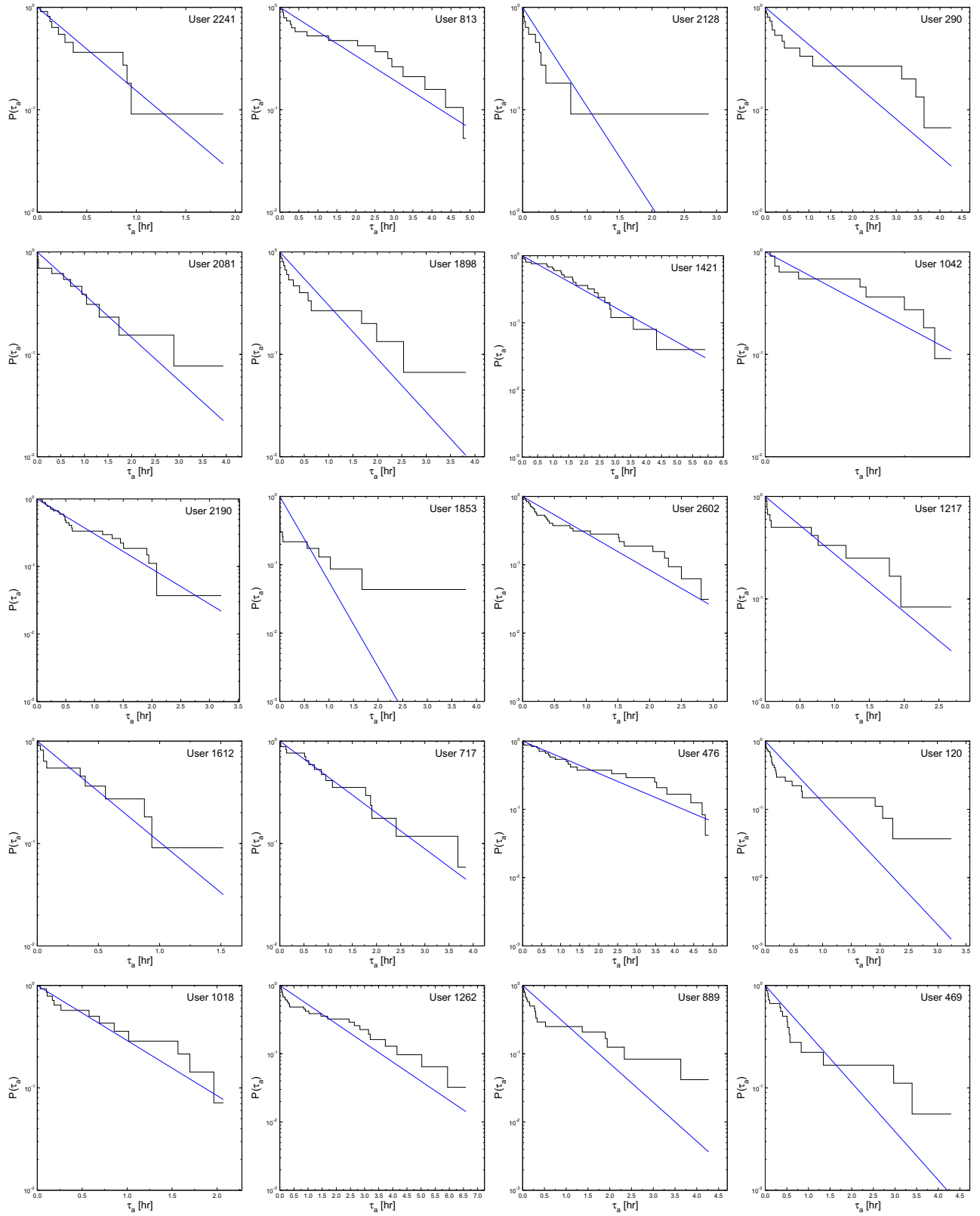


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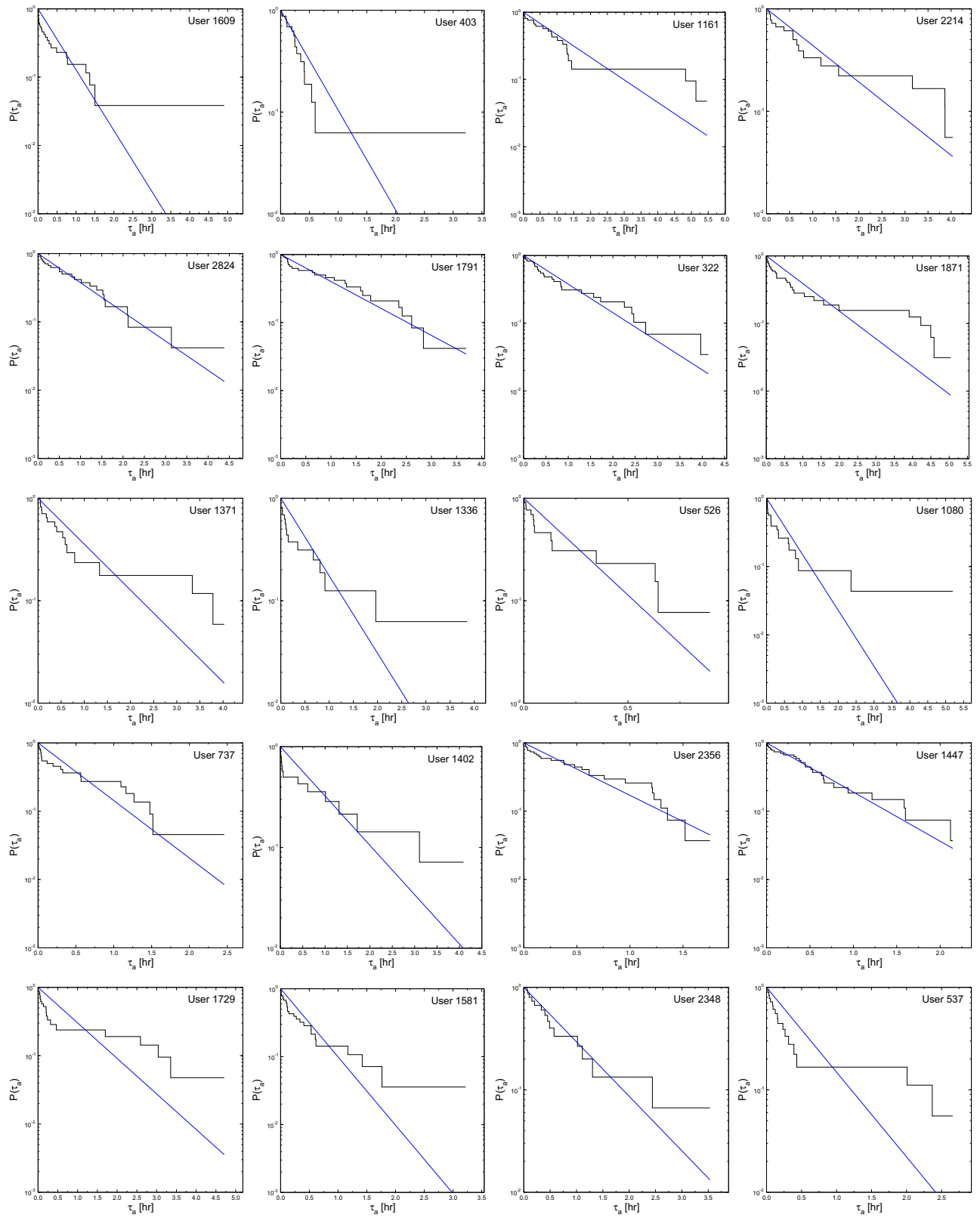


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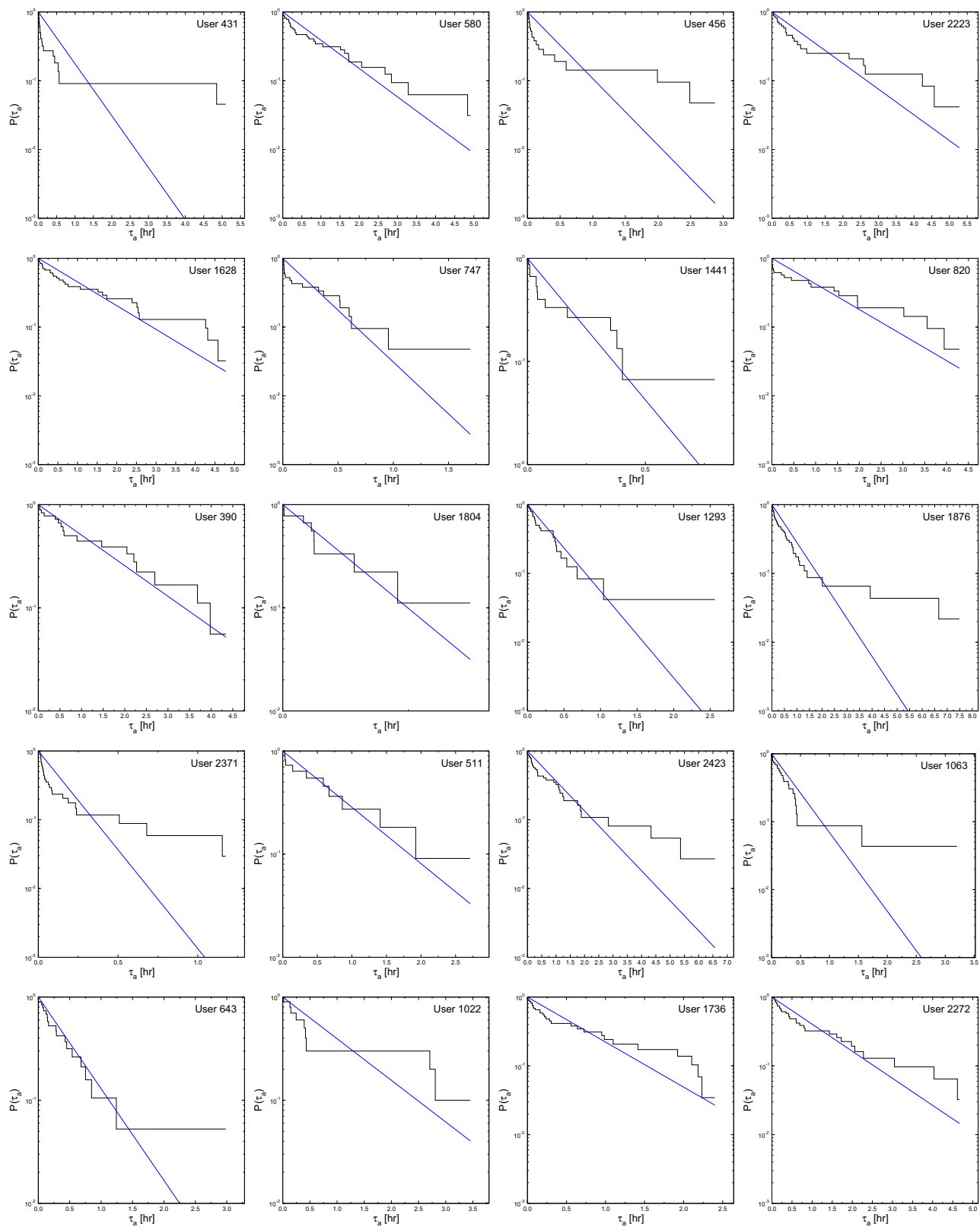


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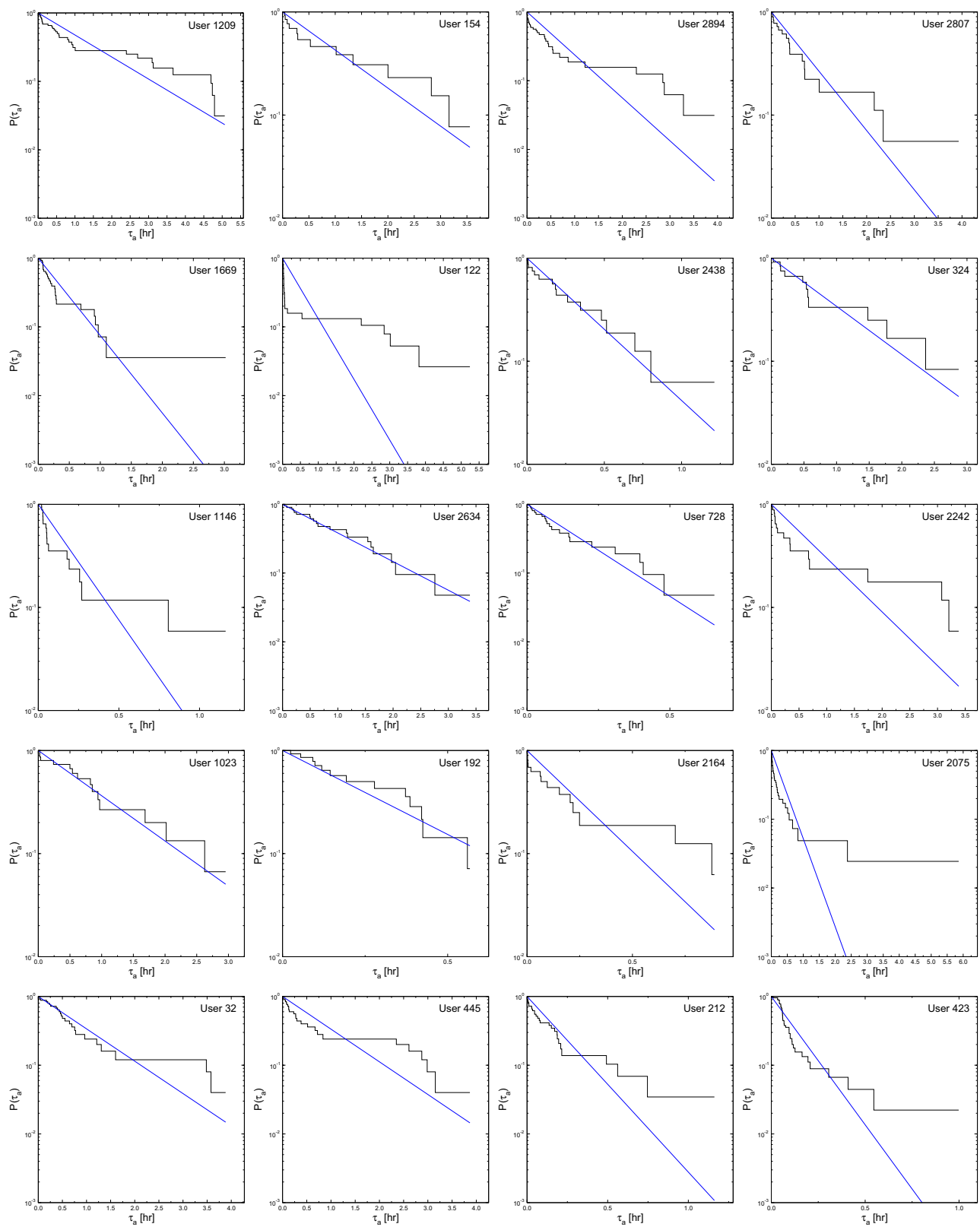


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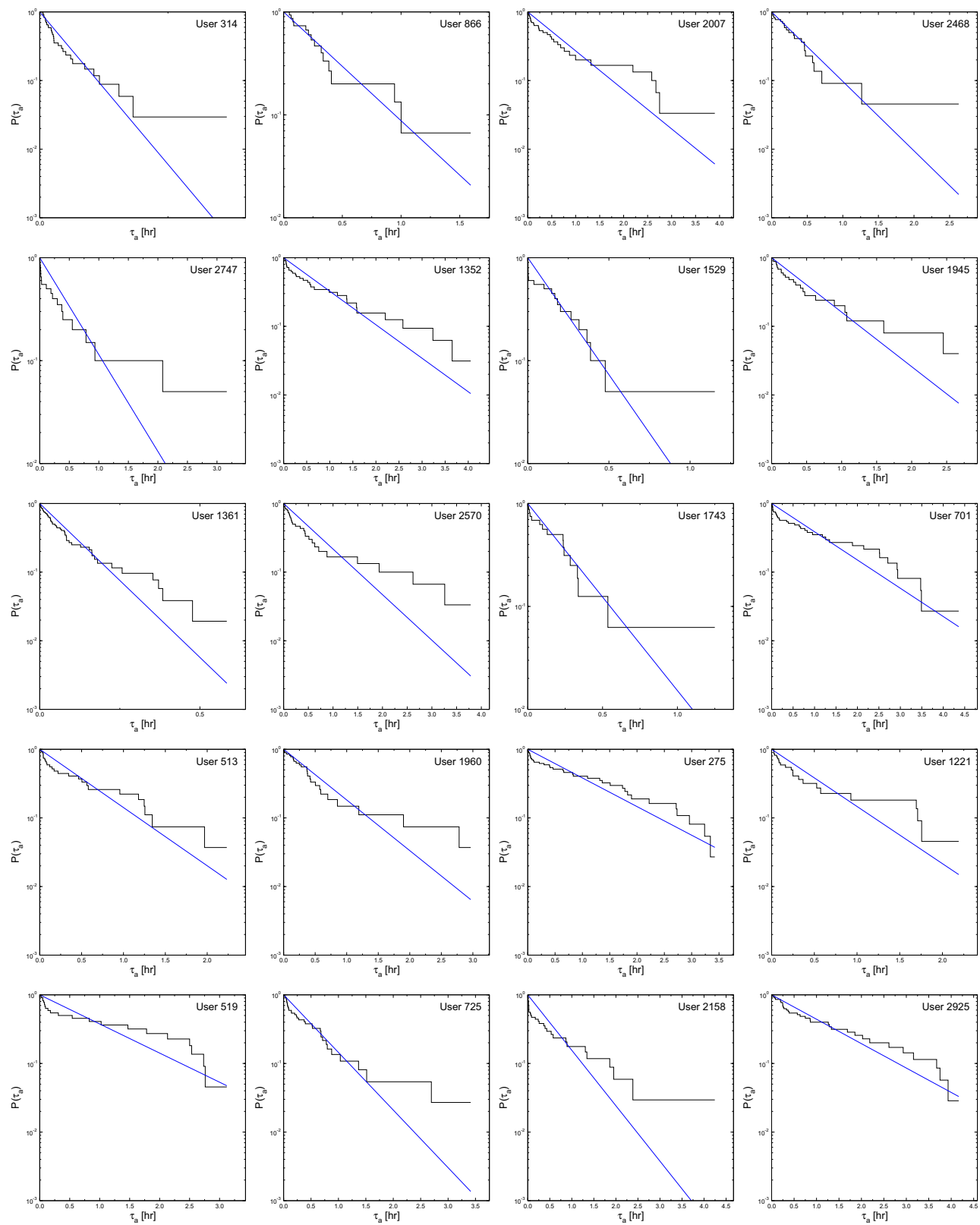


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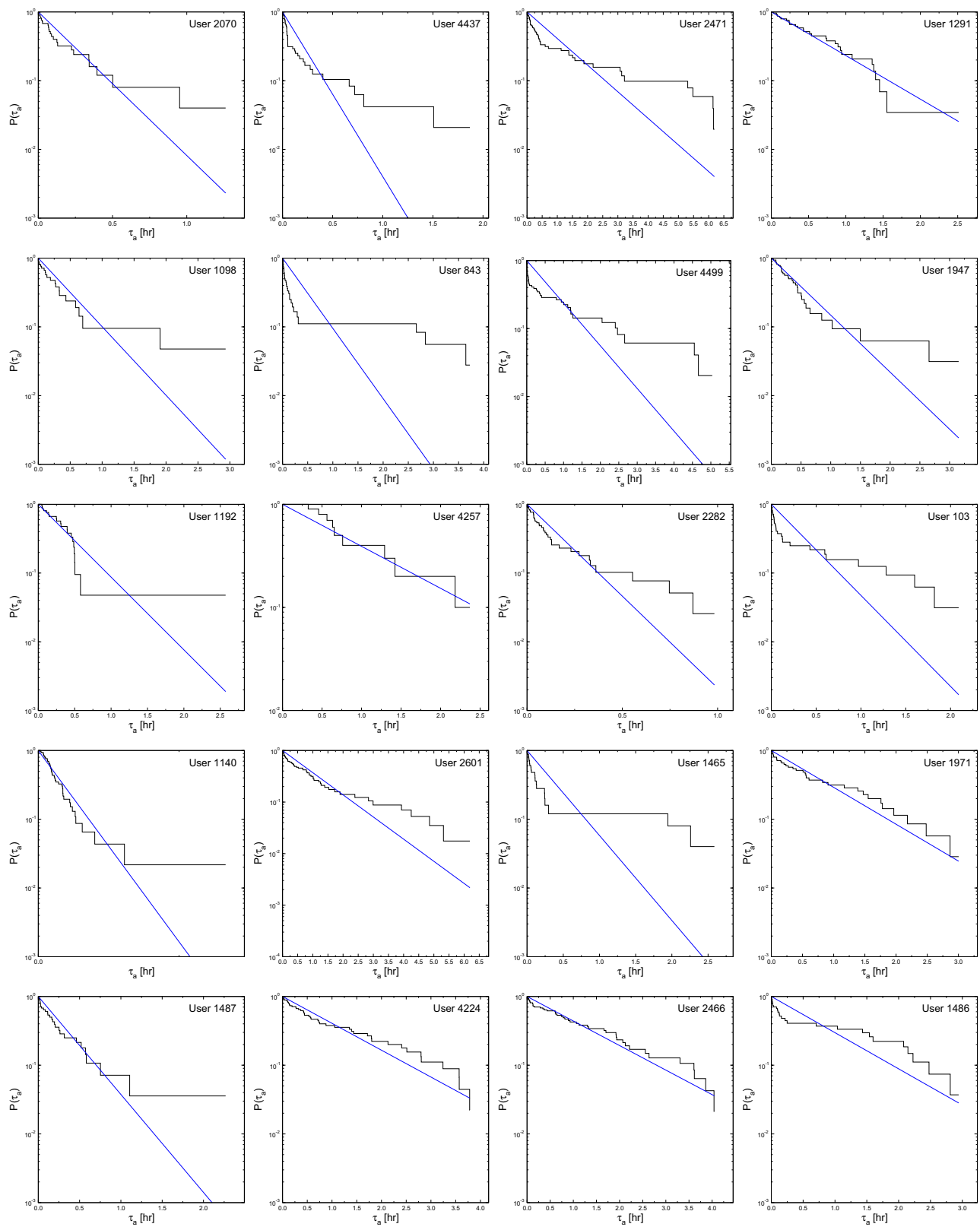


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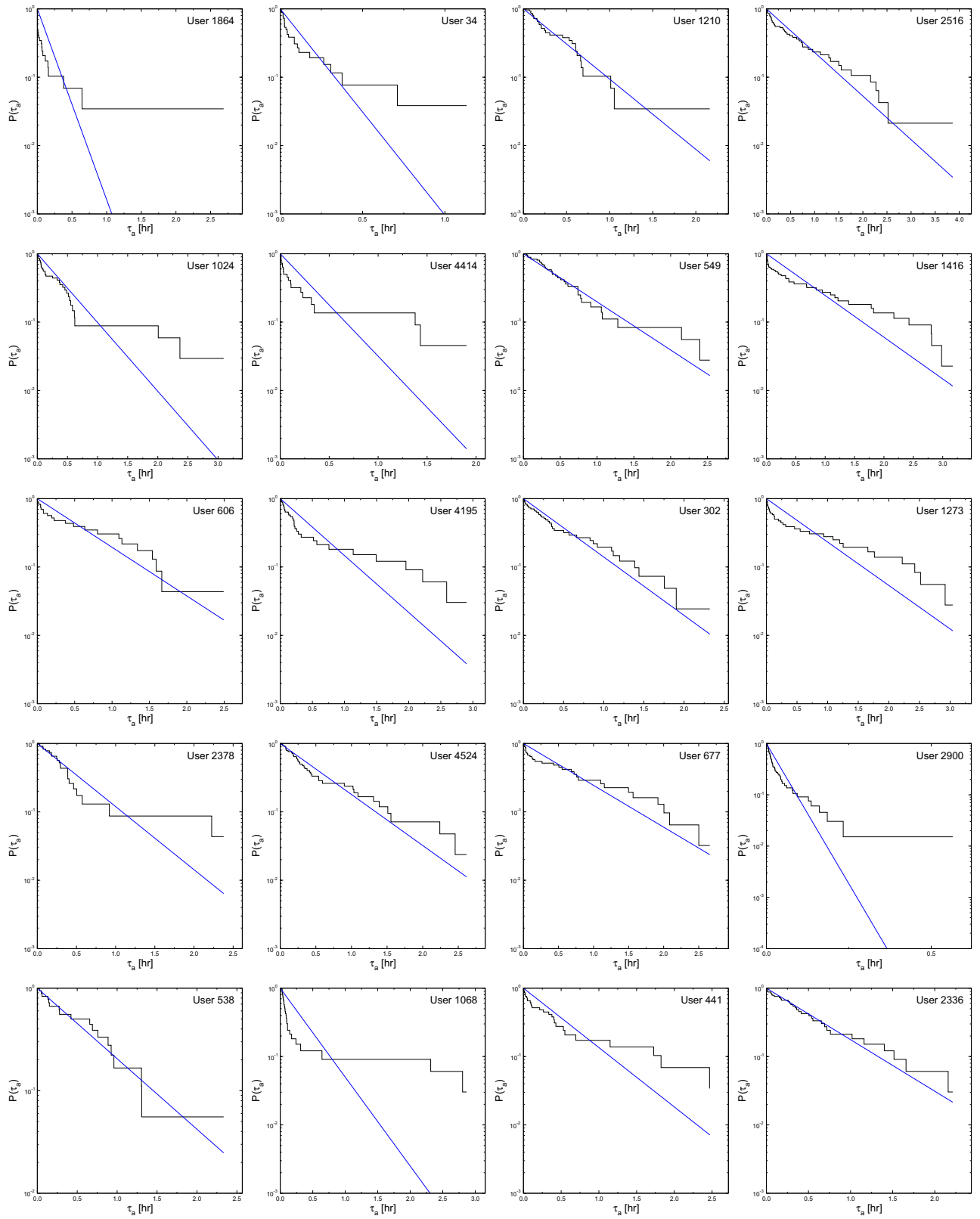


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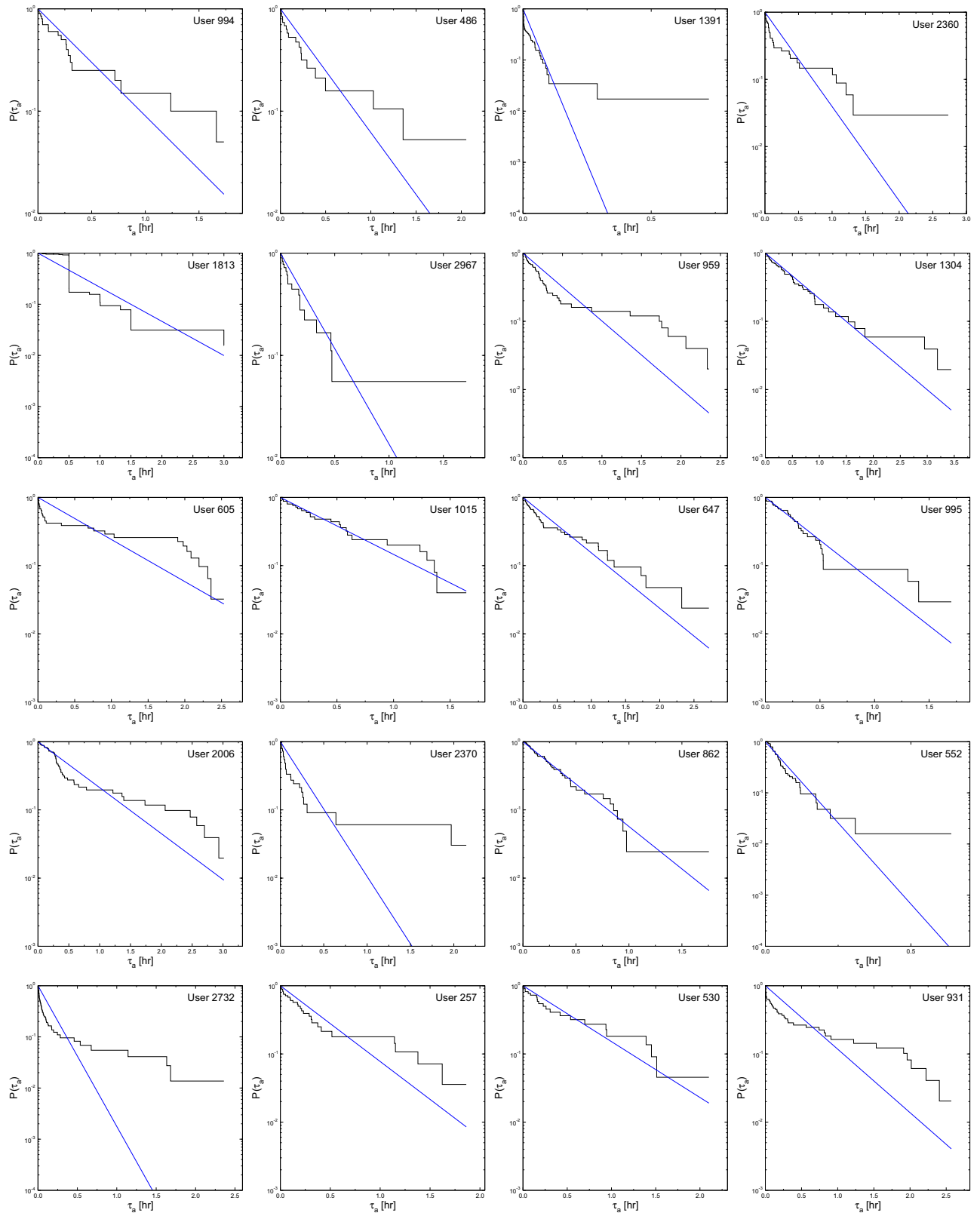


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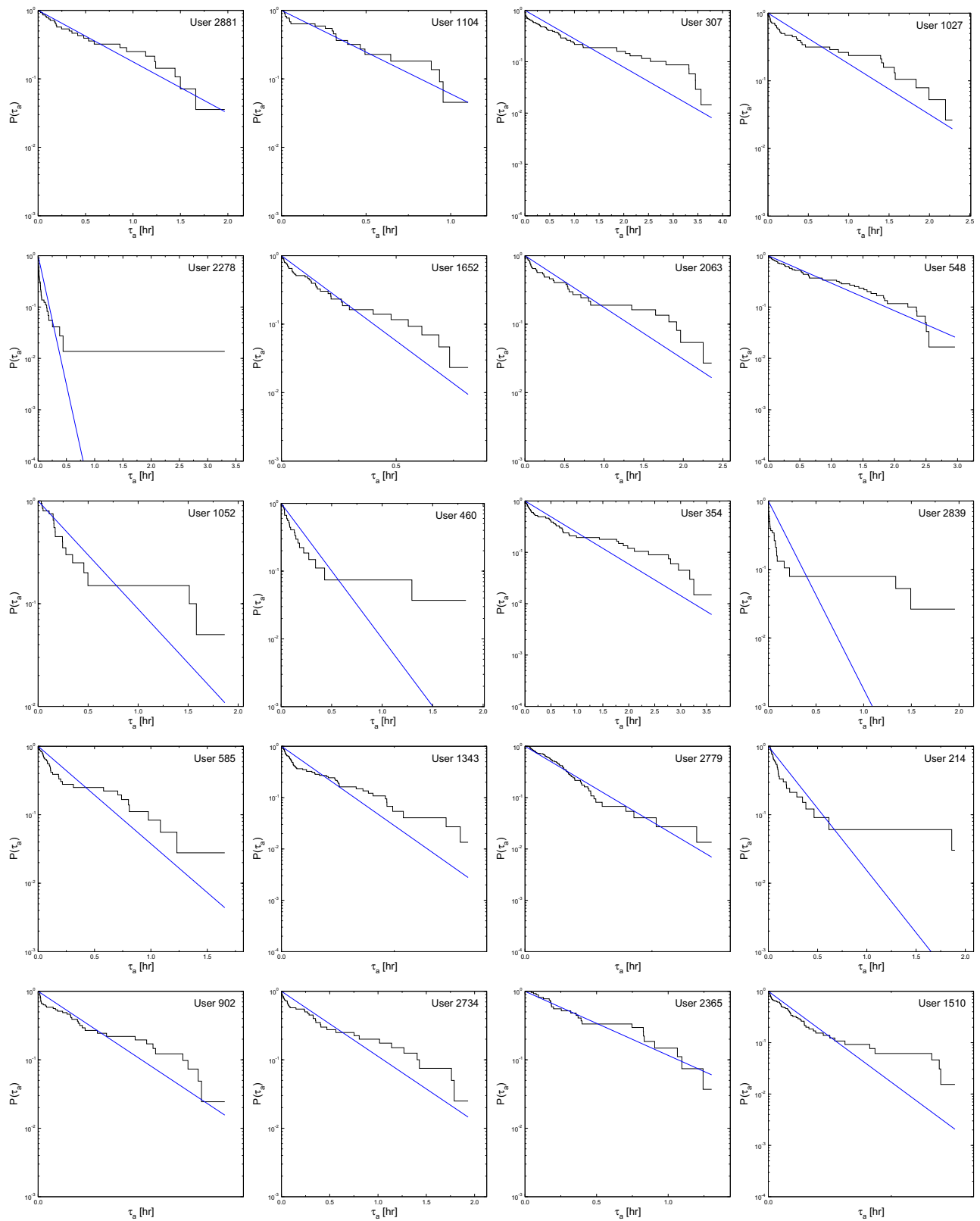


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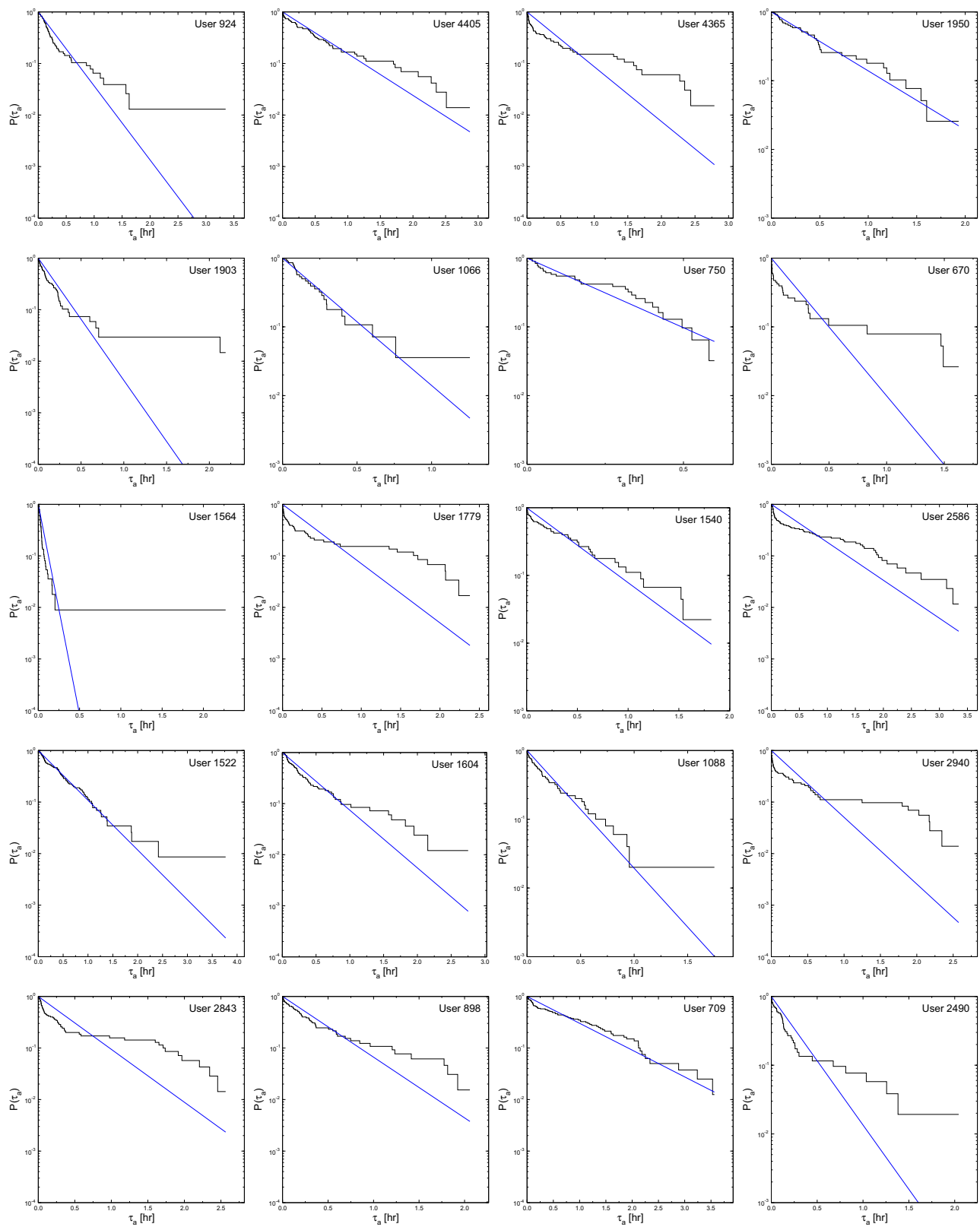


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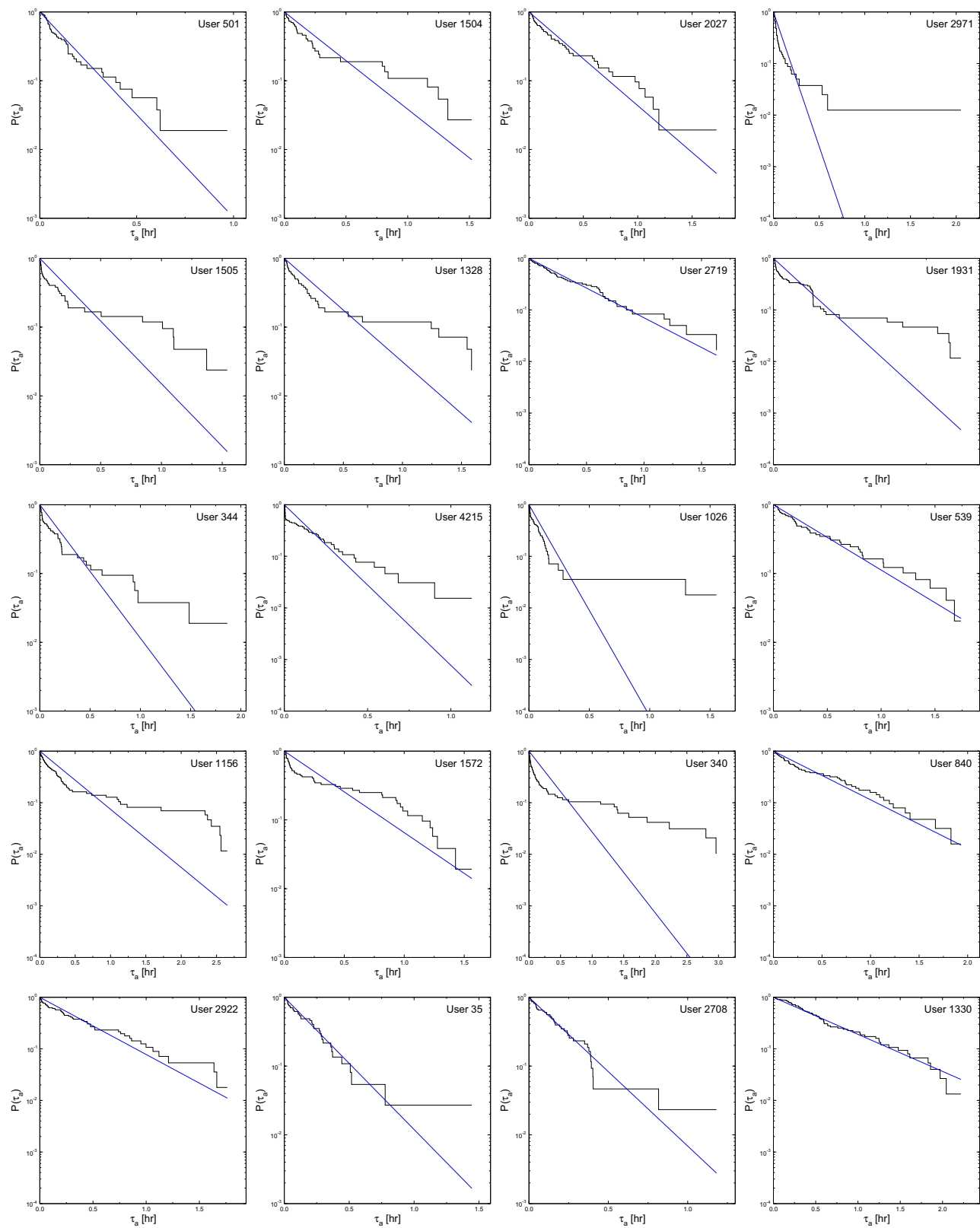


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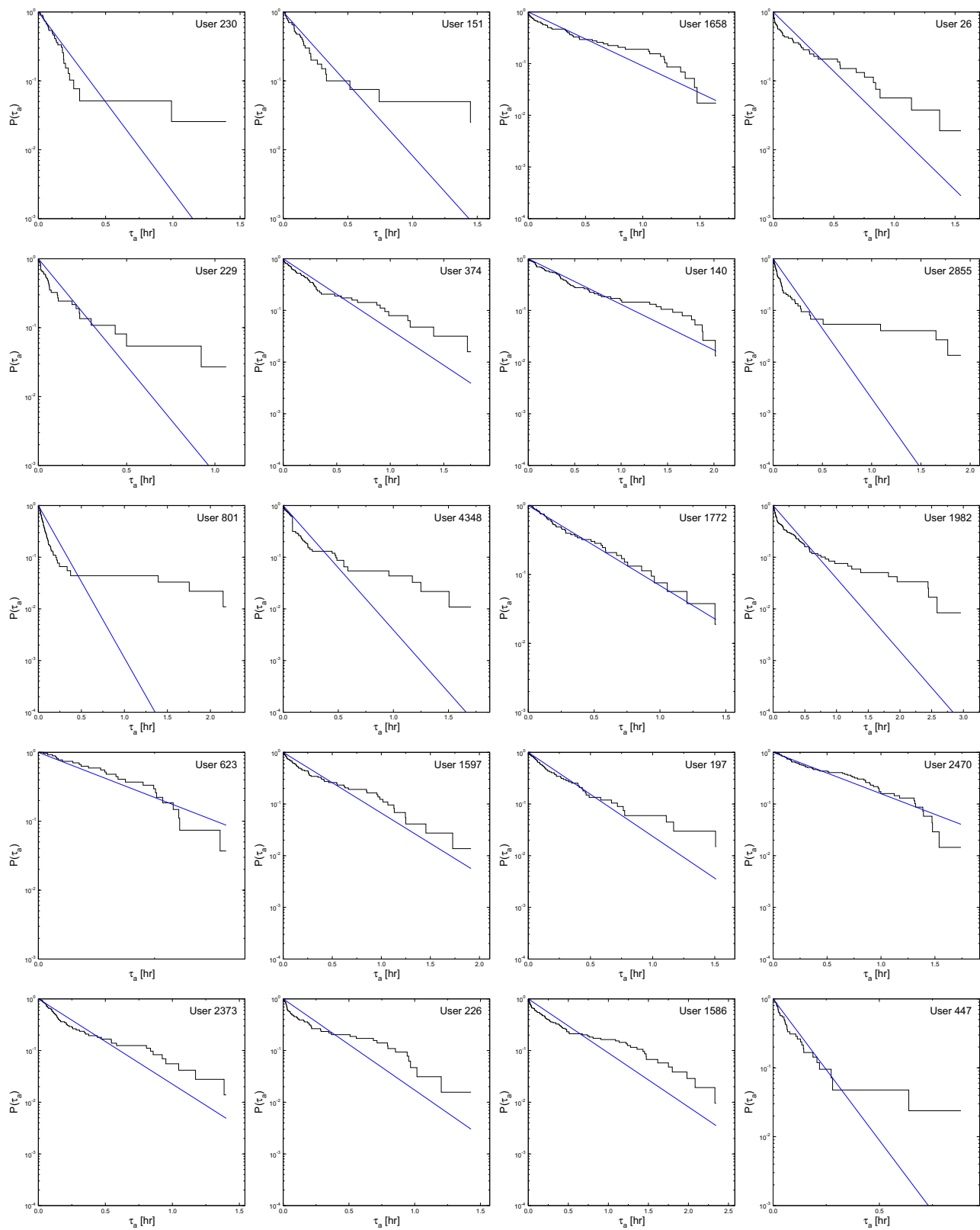


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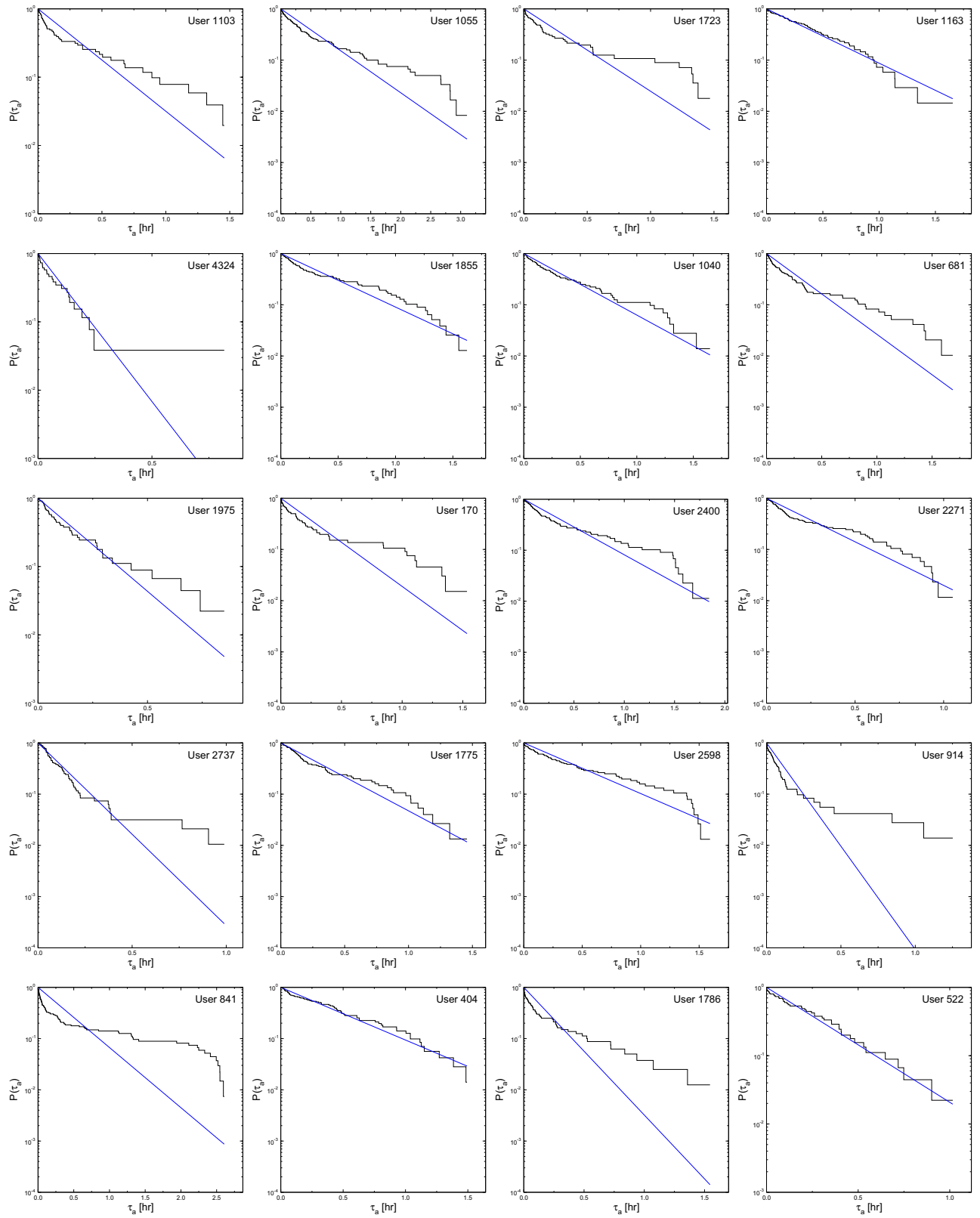


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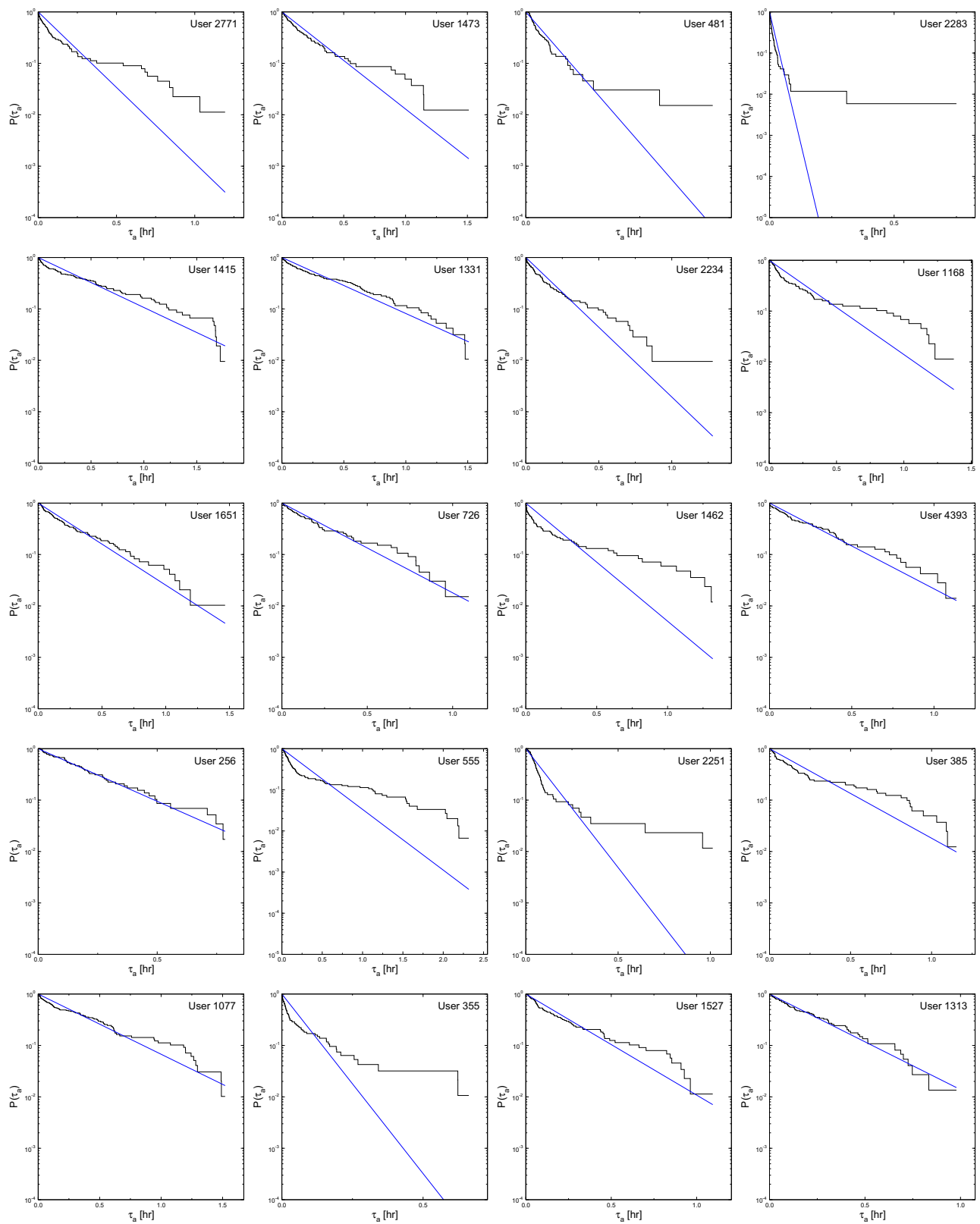


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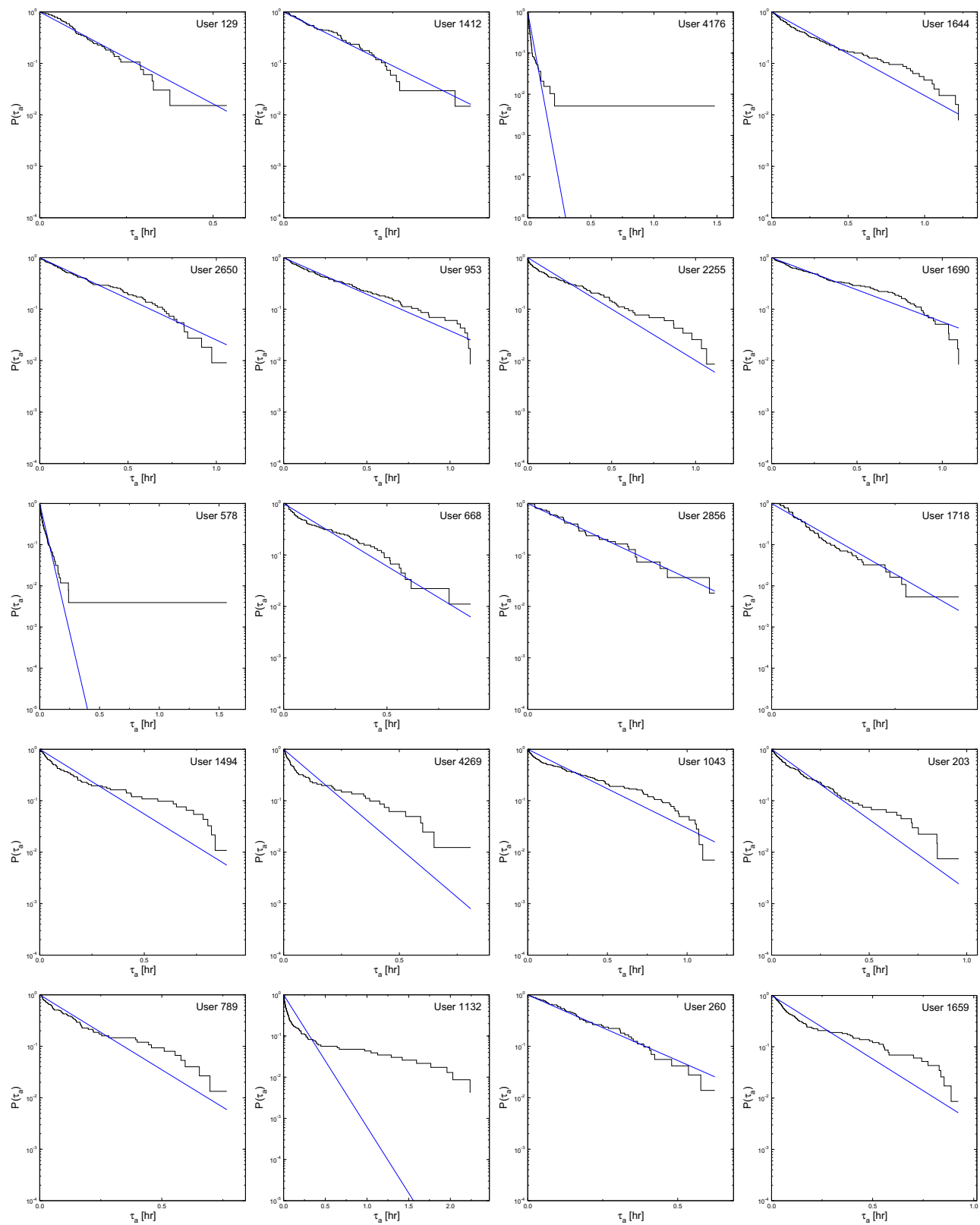


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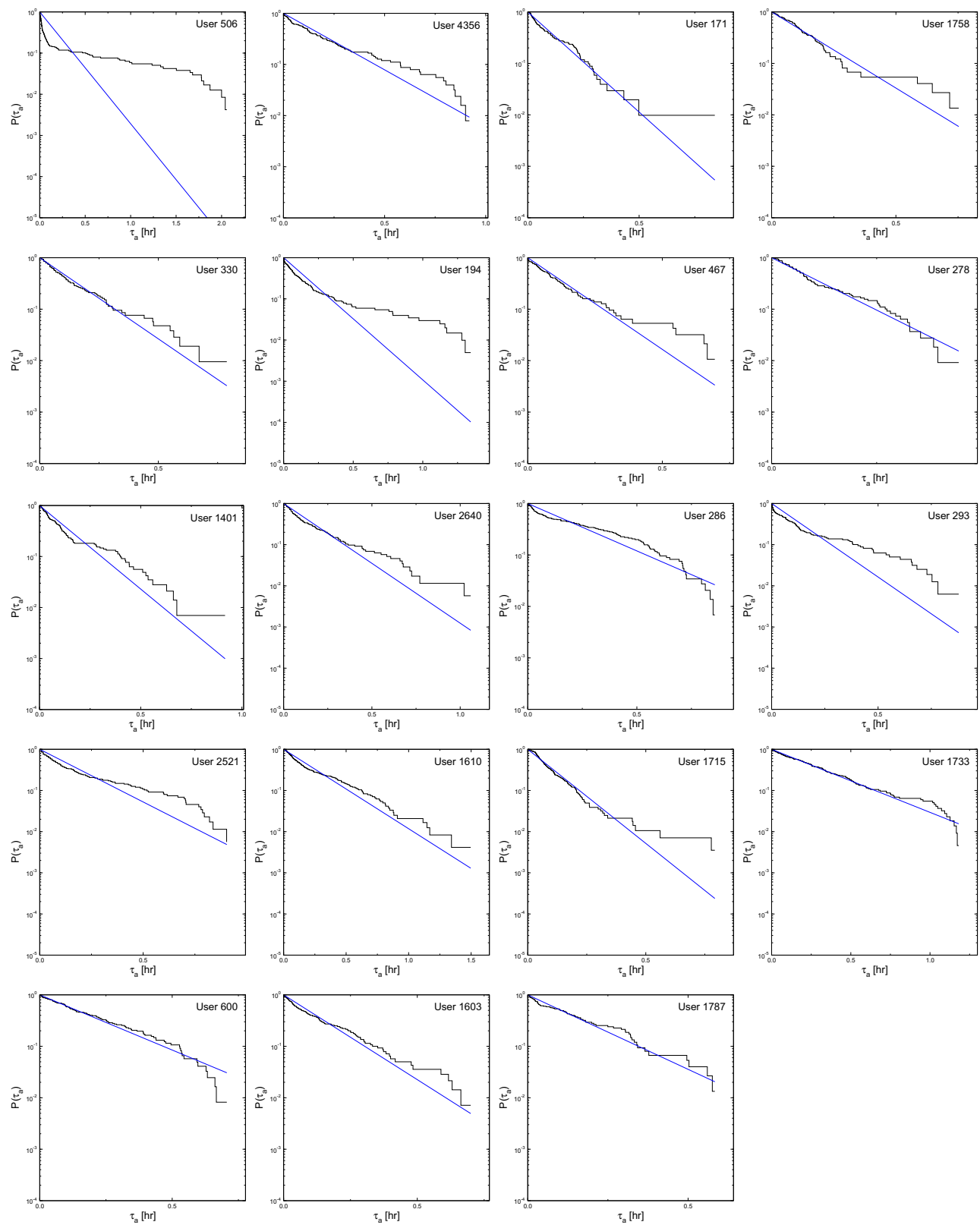


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Other Supporting Information Files

[Appendix \(PDF\)](#)